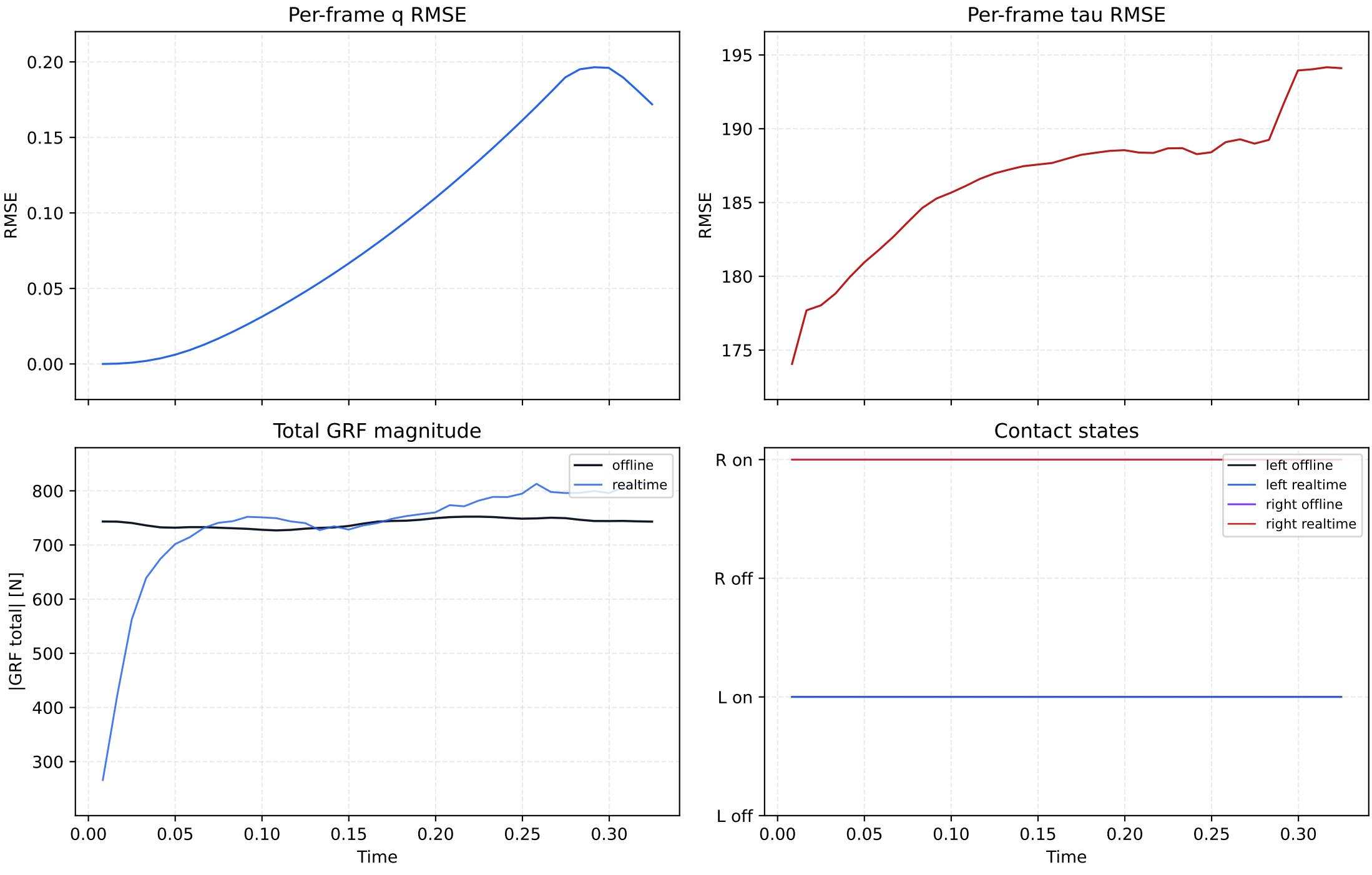
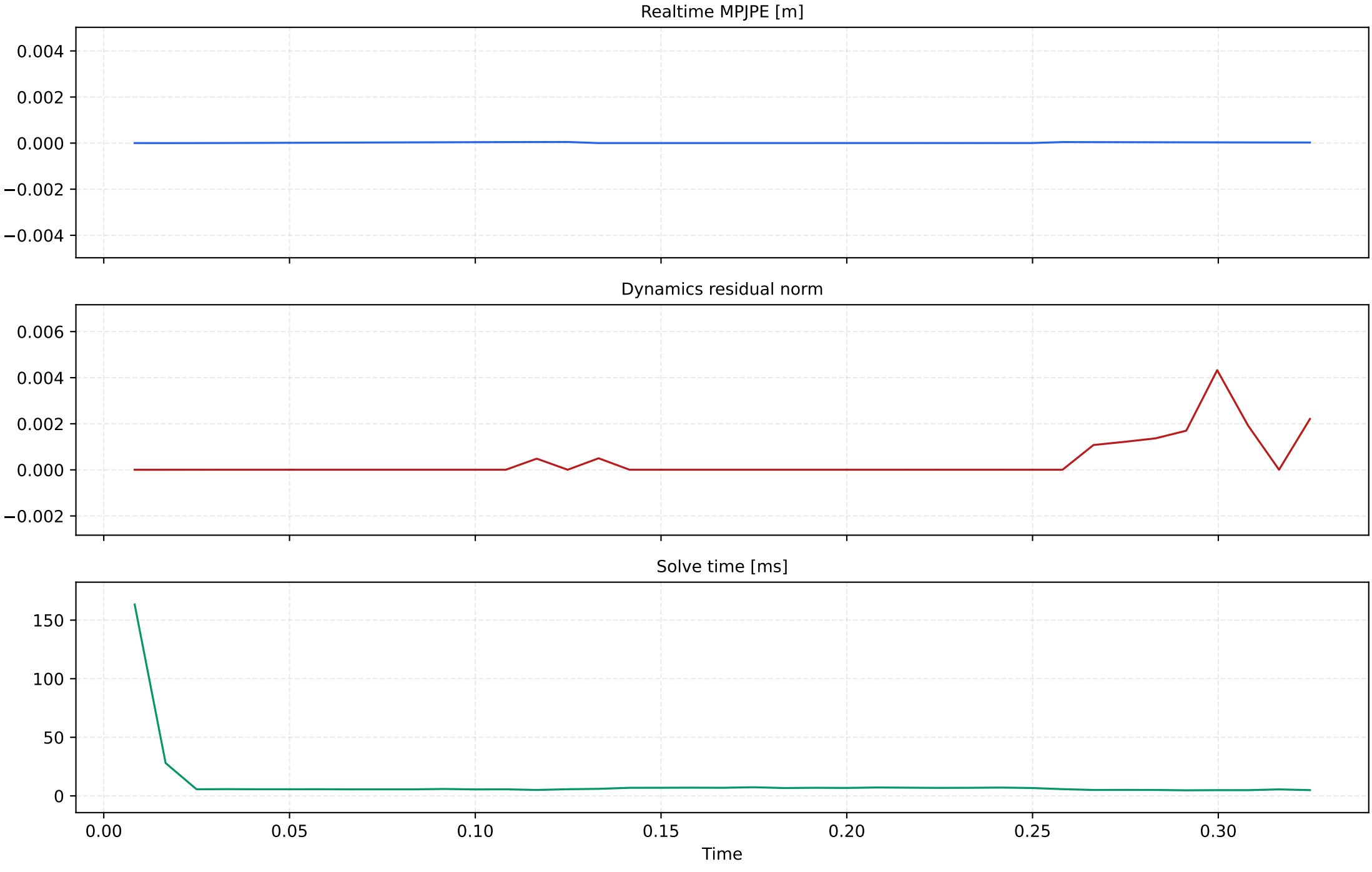
Realtime vs Offline CSV Report

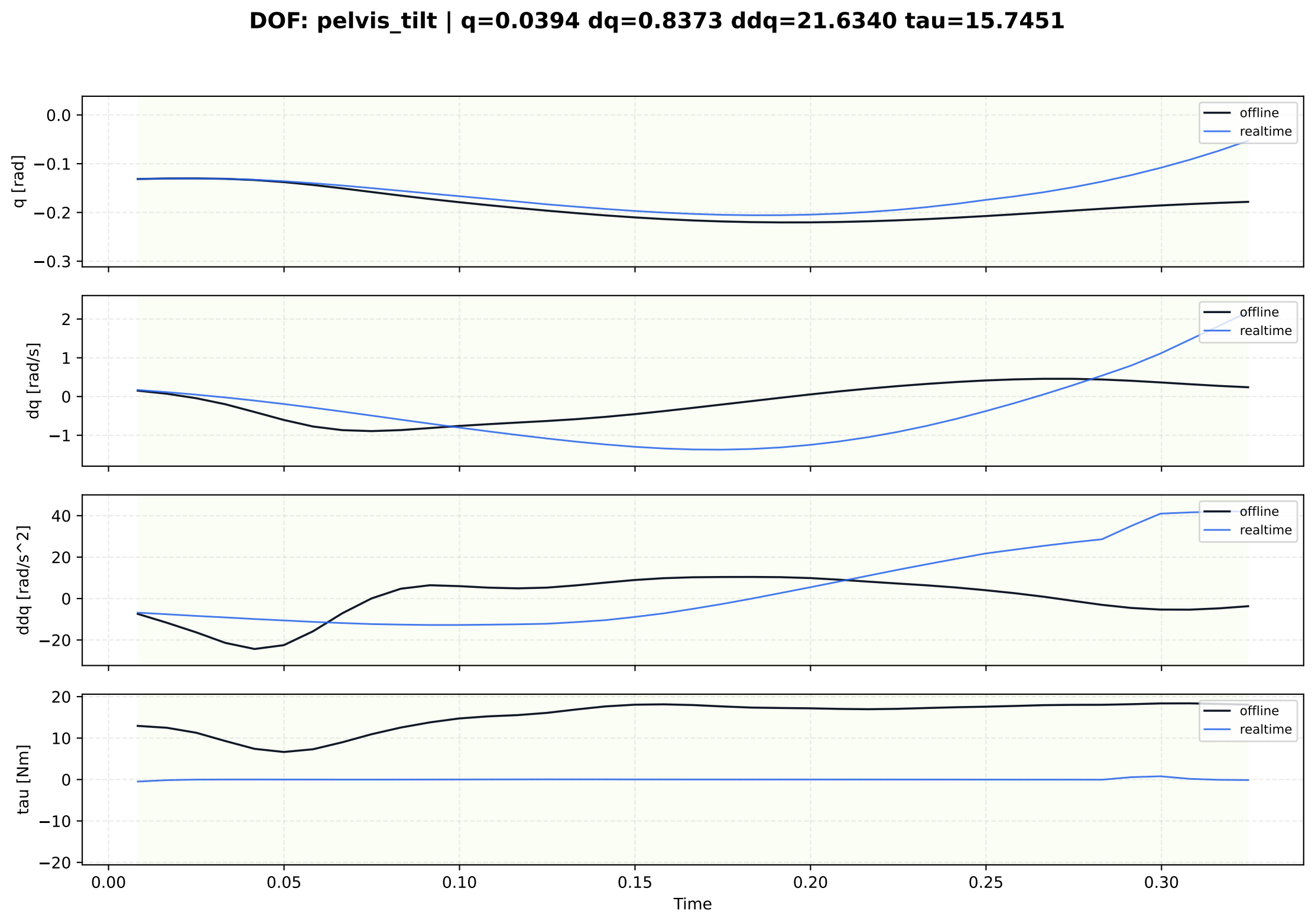
```
Offline CSV:  /Users/enricomartini/Desktop/opensim-batch-dynamics/data/AMASS/BMLhandball/Trial_upper_left_012_poses.csv
Realtime CSV: /Users/enricomartini/Desktop/opensim-batch-dynamics/RT/tmp_rt_sequence.csv

Matched frames:       39
Alignment key:        frame
Common DOFs:          49

q RMSE:               0.113013
q MAE:                0.037949
dq RMSE:              1.024080
ddq RMSE:             33.420335
tau full RMSE:        186.716550
tau actuated RMSE:    118.123727
left GRF RMSE:        101.862462
right GRF RMSE:       104.189836

left contact F1:      1.0000
right contact F1:     1.0000

realtime MPJPE mean:  0.000019 m
realtime MPJPE max:   0.000049 m
dyn residual mean:    0.000384
solve time mean [ms]: 10.611
solve time p95  [ms]: 9.442
```

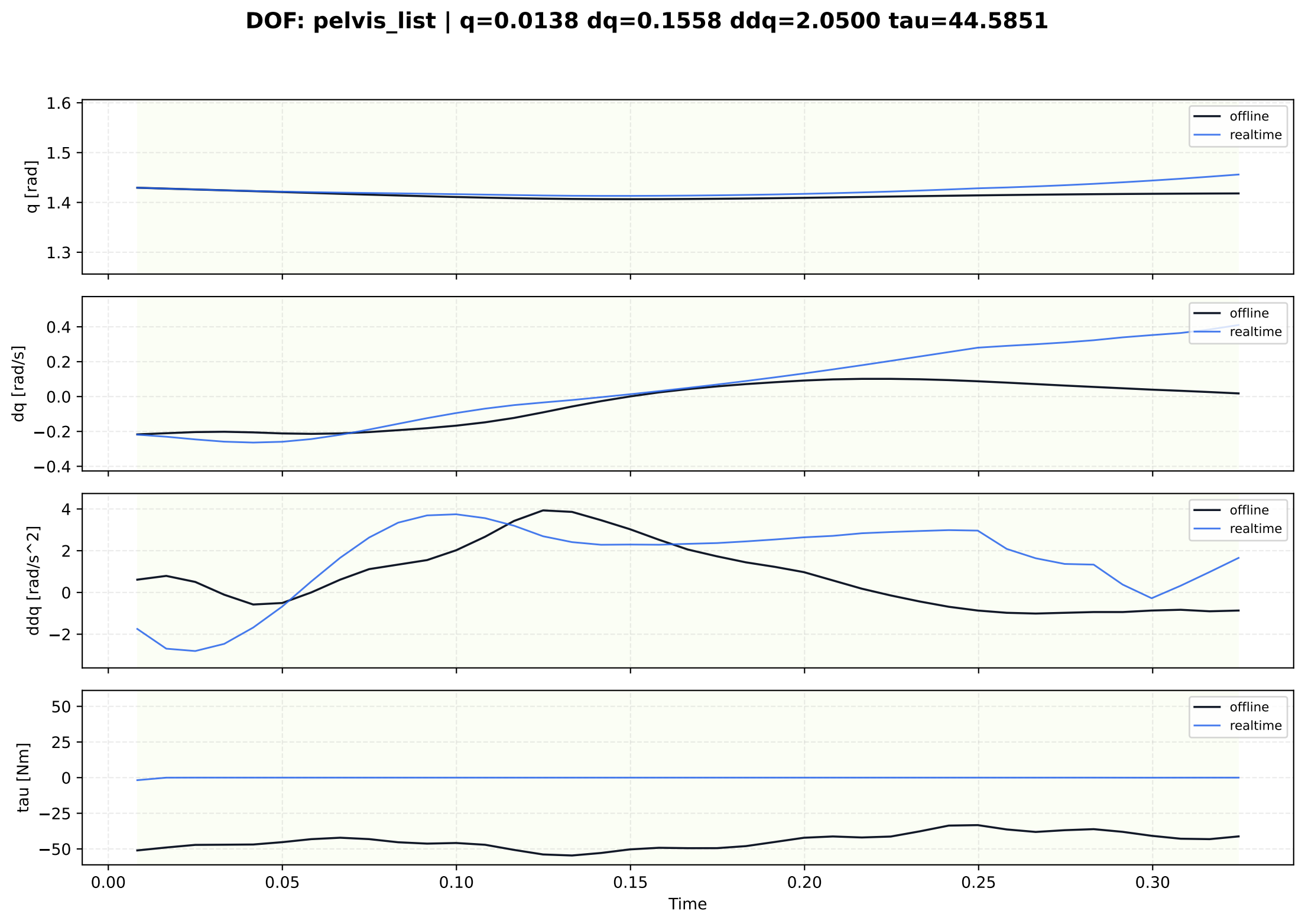
Overview

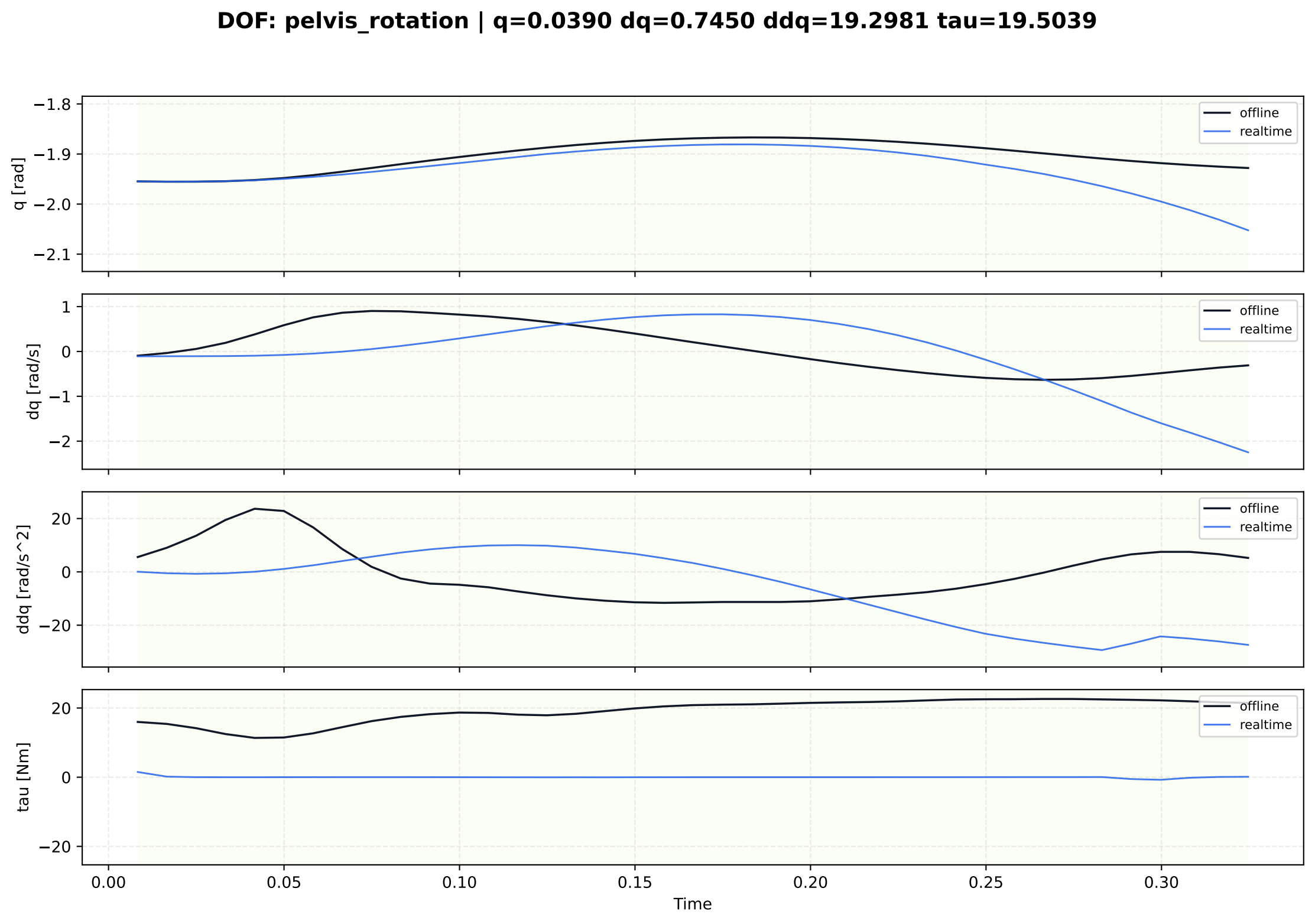

Per-frame q RMSE
RMSE
0.20
0.15
0.10
0.05
0.00
Per-frame tau RMSE
RMSE
195
190
185
180
175
Total GRF magnitude
|GRF total| [N]
800
700
600
500
400
300
offline
realtime
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Time
Contact states
R on
R off
L on
L off
left offline
left realtime
right offline
right realtime
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Time

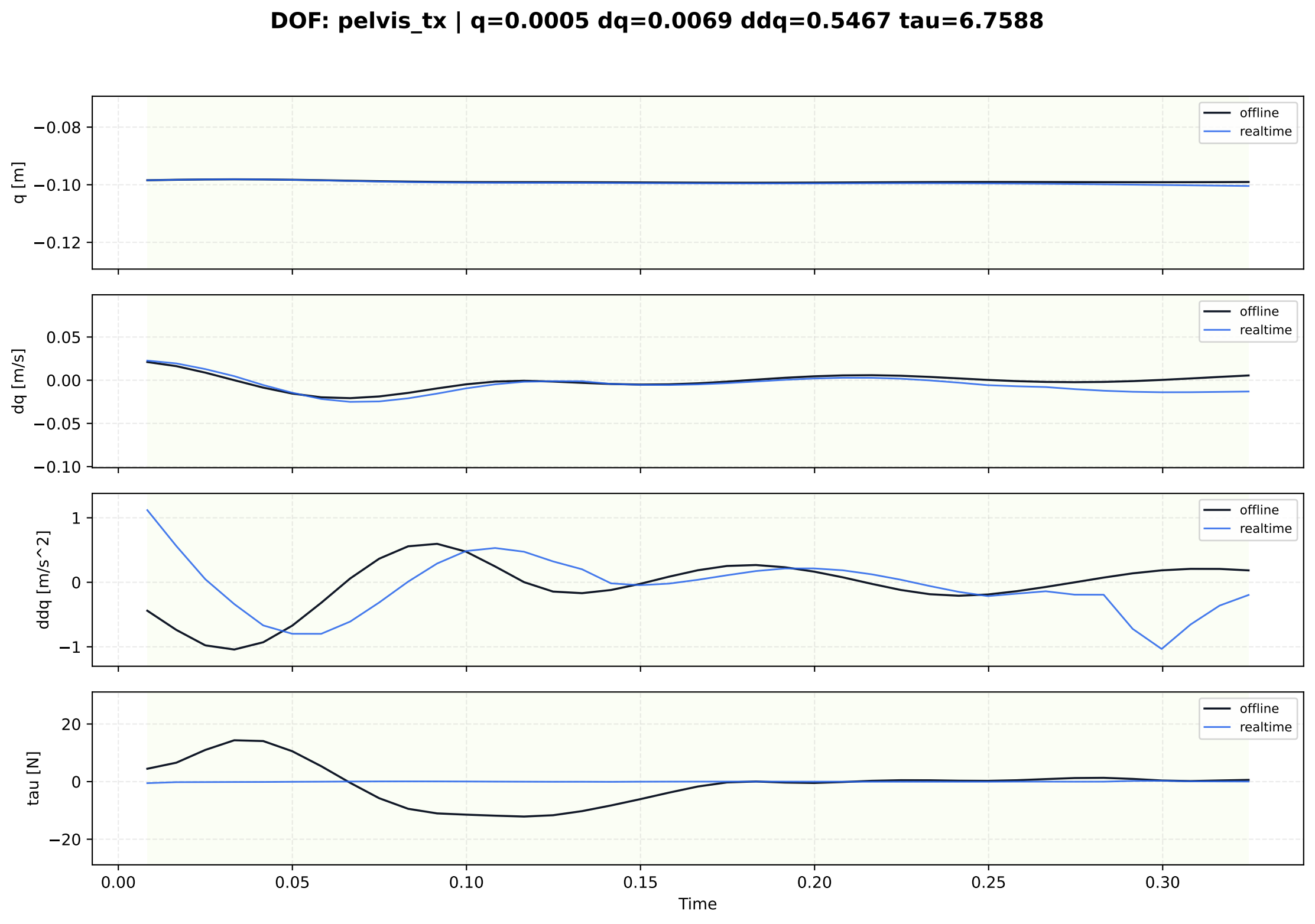
Realtime Diagnostics

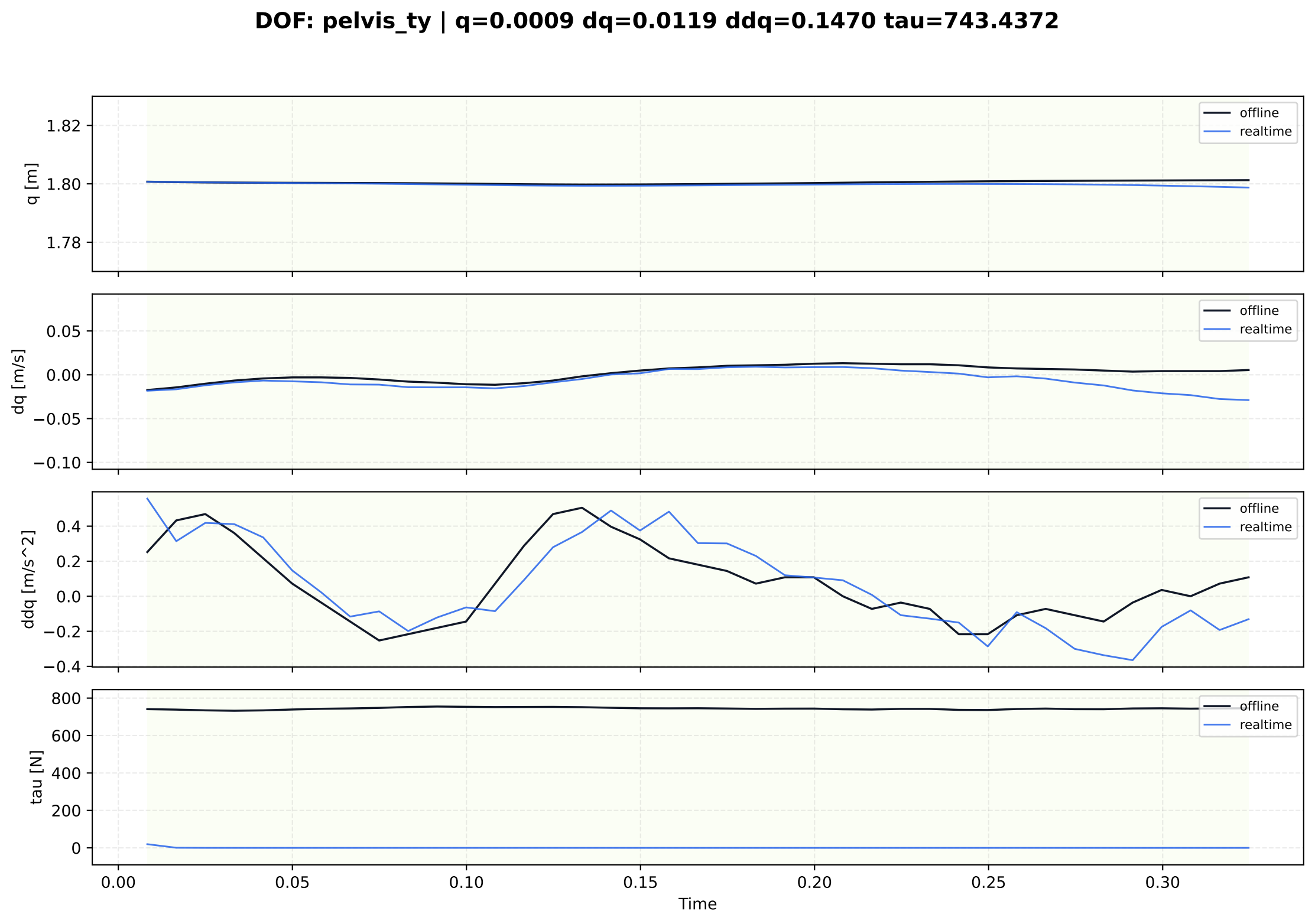


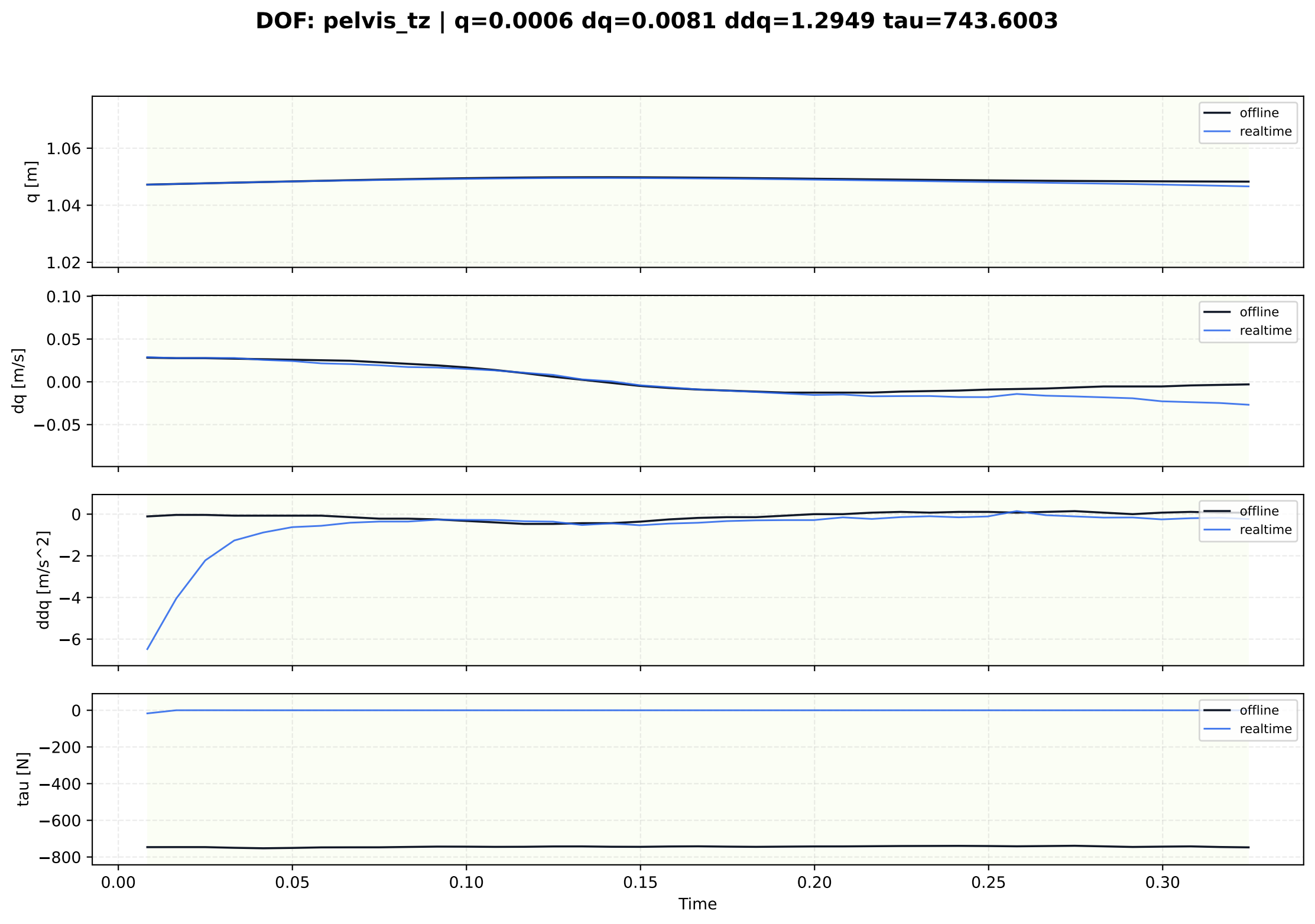

DOF: pelvis_tilt | q=0.0394 dq=0.8373 ddq=21.6340 tau=15.7451
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

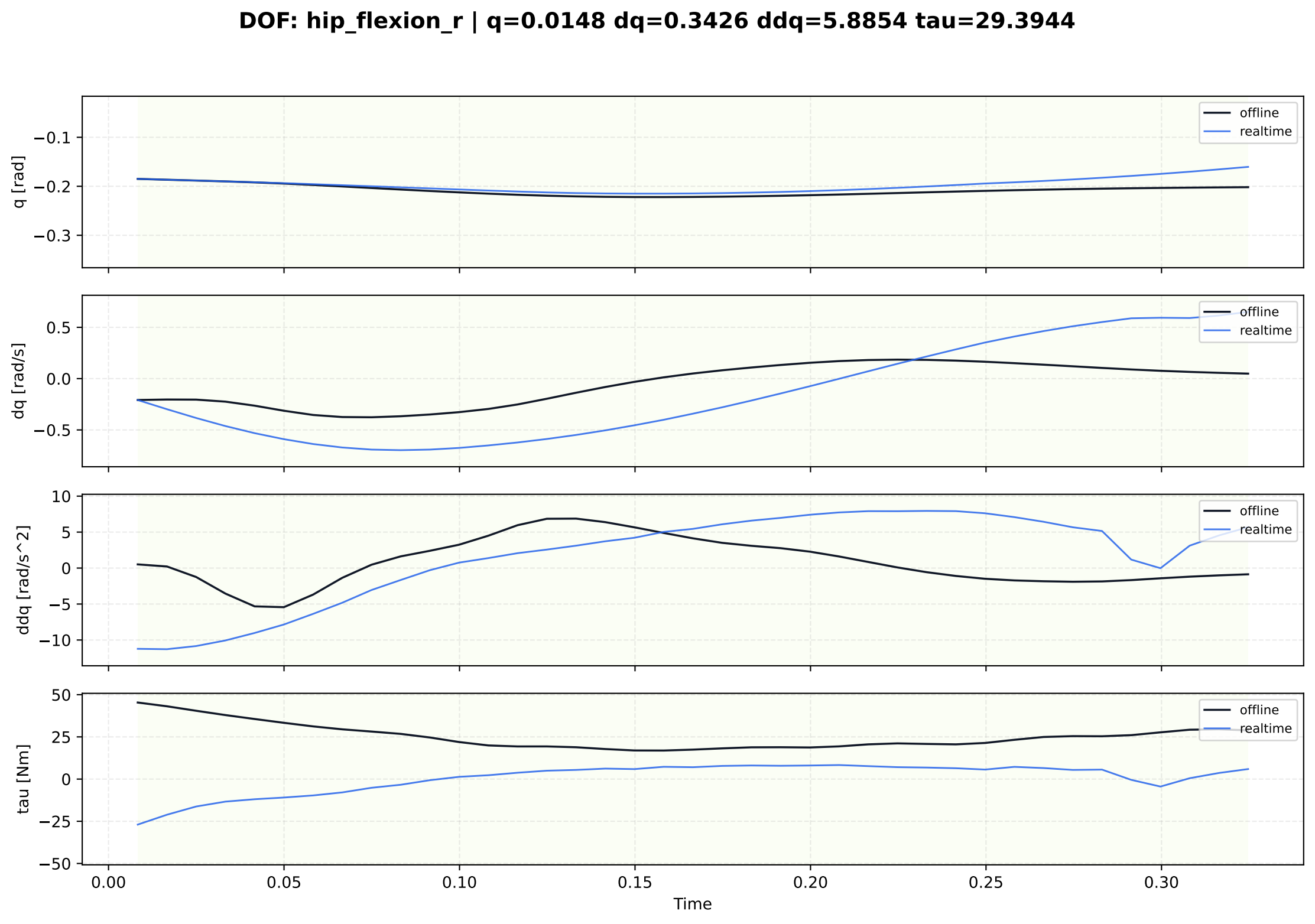

DOF: pelvis_list | q=0.0138 dq=0.1558 ddq=2.0500 tau=44.5851
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

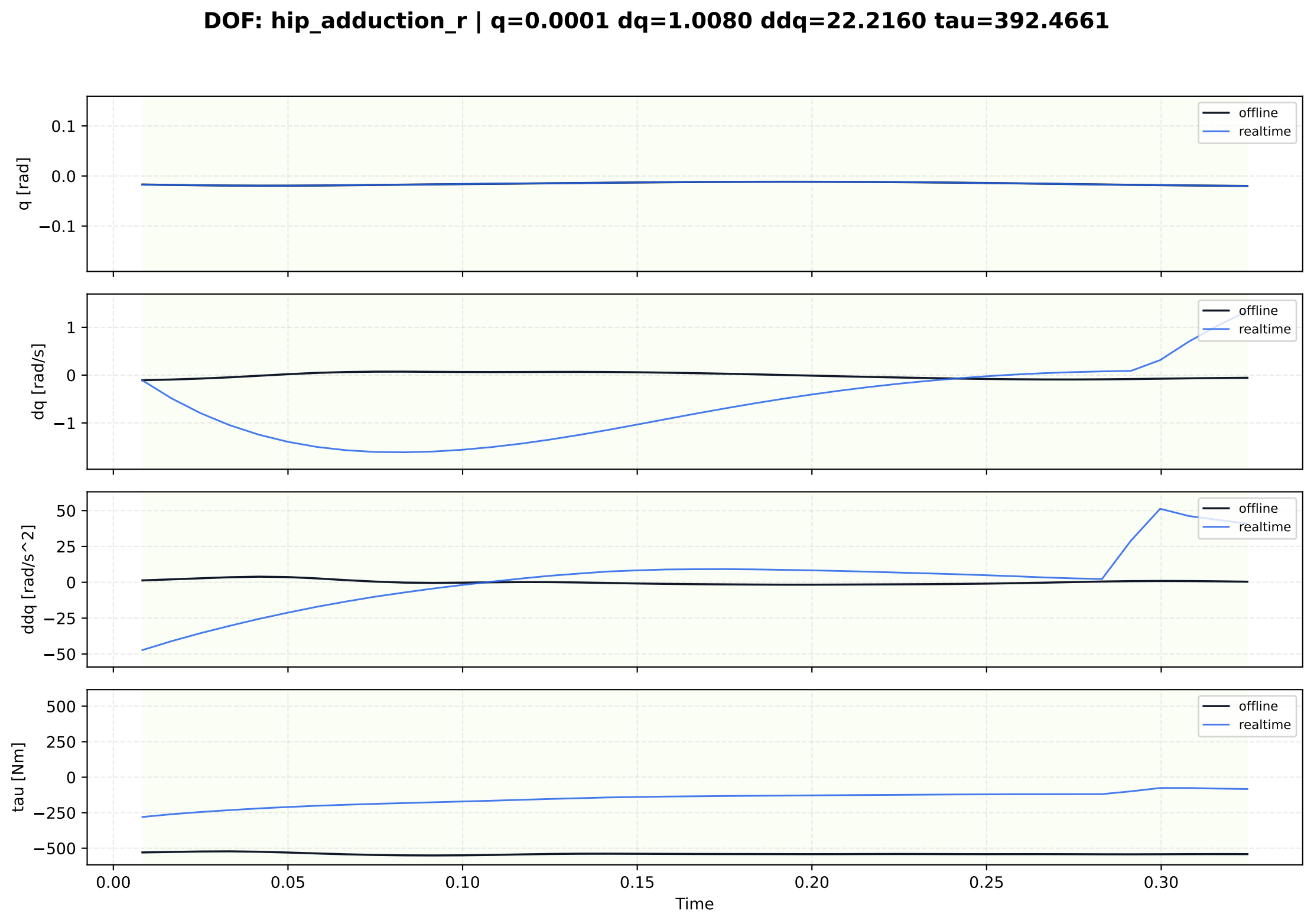

DOF: pelvis_rotation | q=0.0390 dq=0.7450 ddq=19.2981 tau=19.5039
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
Time
offline
realtime

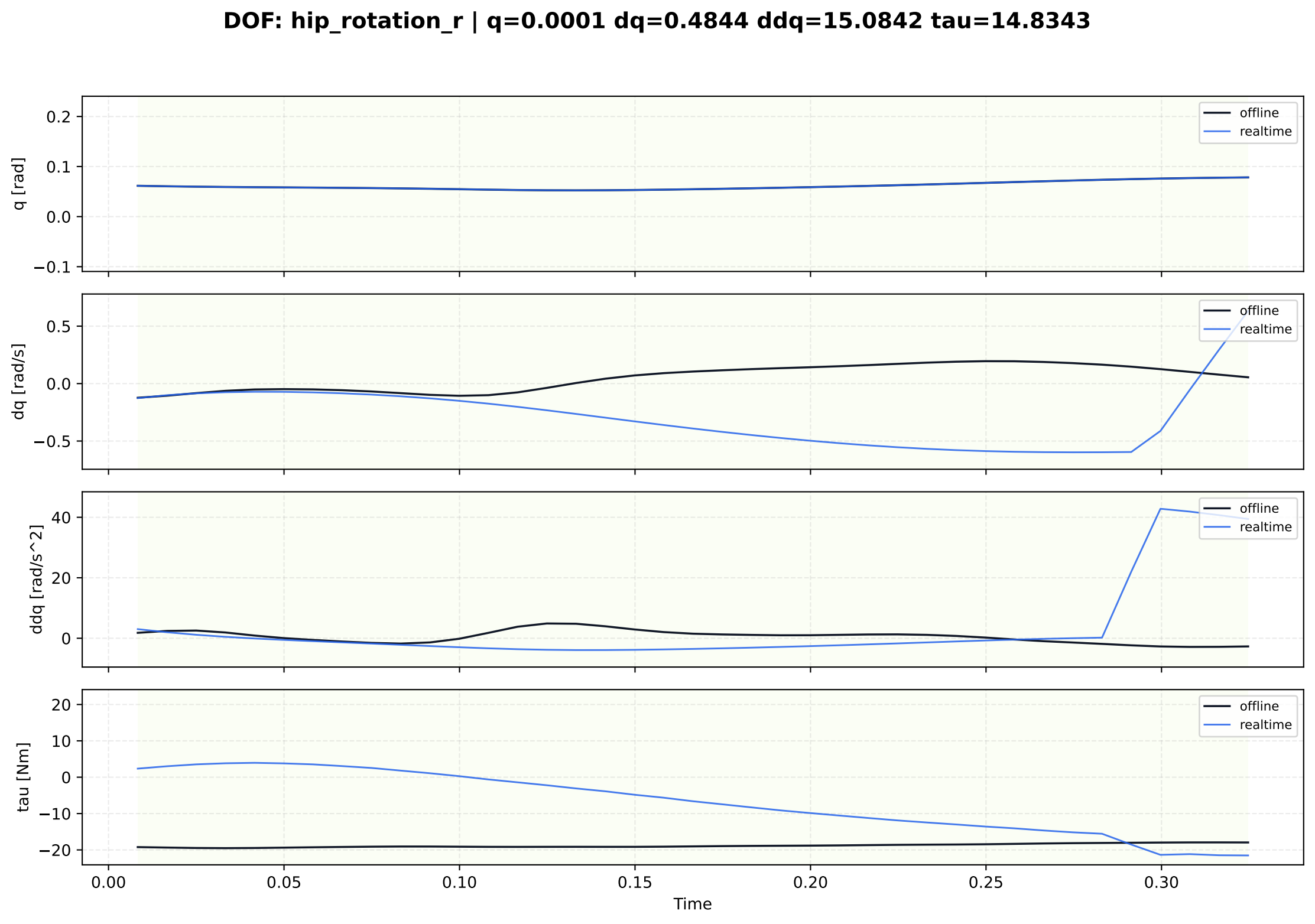

DOF: pelvis_tx | q=0.0005 dq=0.0069 ddq=0.5467 tau=6.7588
q [m]
dq [m/s]
ddq [m/s^2]
tau [N]
offline
realtime
Time

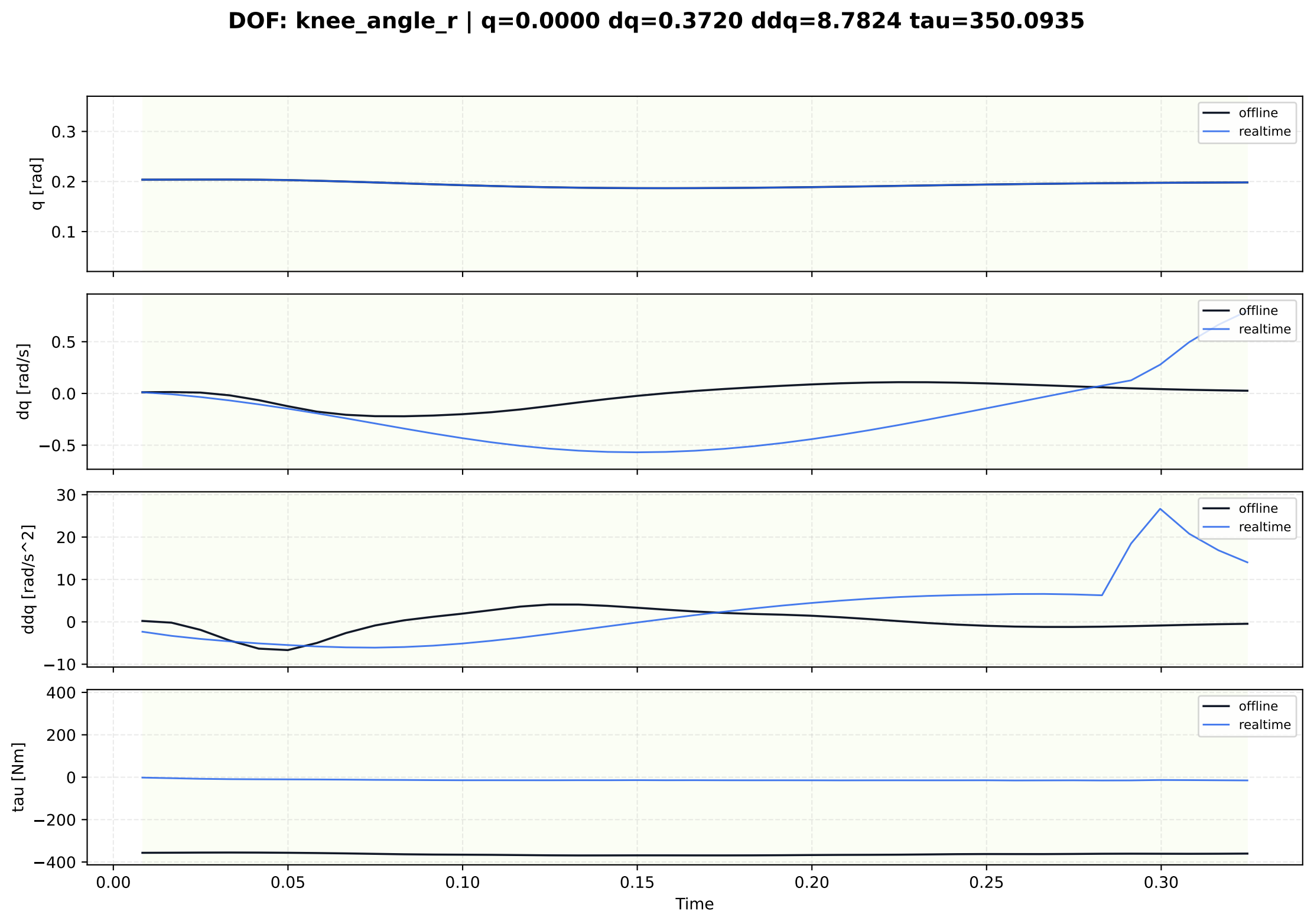

DOF: pelvis_ty | q=0.0009 dq=0.0119 ddq=0.1470 tau=743.4372
q [m]
1.82
1.80
1.78
dq [m/s]
0.05
0.00
−0.05
−0.10
ddq [m/s^2]
0.4
0.2
0.0
−0.2
−0.4
tau [N]
800
600
400
200
0
offline
realtime
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Time

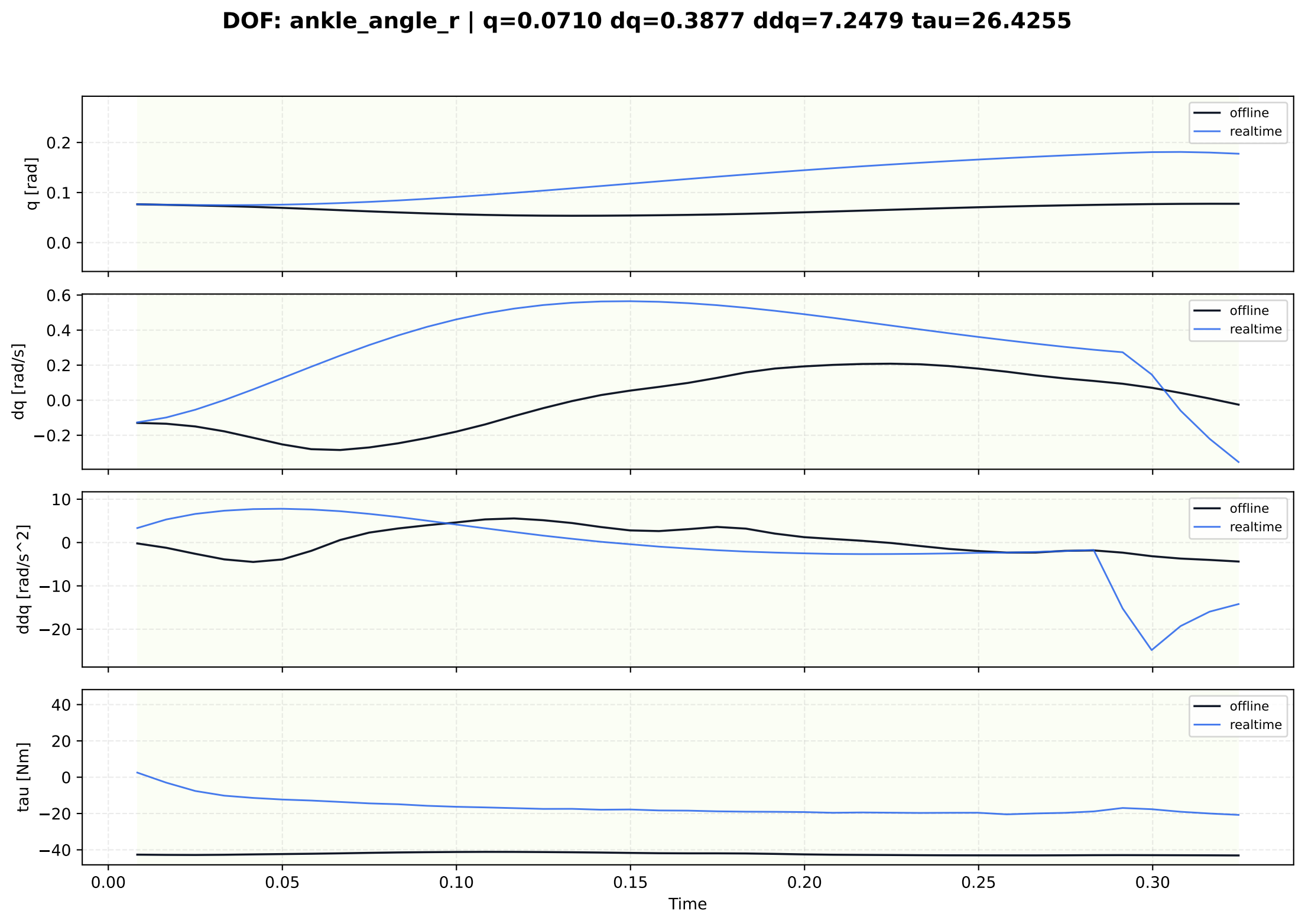

DOF: pelvis_tz | q=0.0006 dq=0.0081 ddq=1.2949 tau=743.6003
q [m]
1.06
1.04
1.02
offline
realtime
dq [m/s]
0.10
0.05
0.00
−0.05
offline
realtime
ddq [m/s^2]
0
−2
−4
−6
offline
realtime
tau [N]
0
−200
−400
−600
−800
offline
realtime
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Time

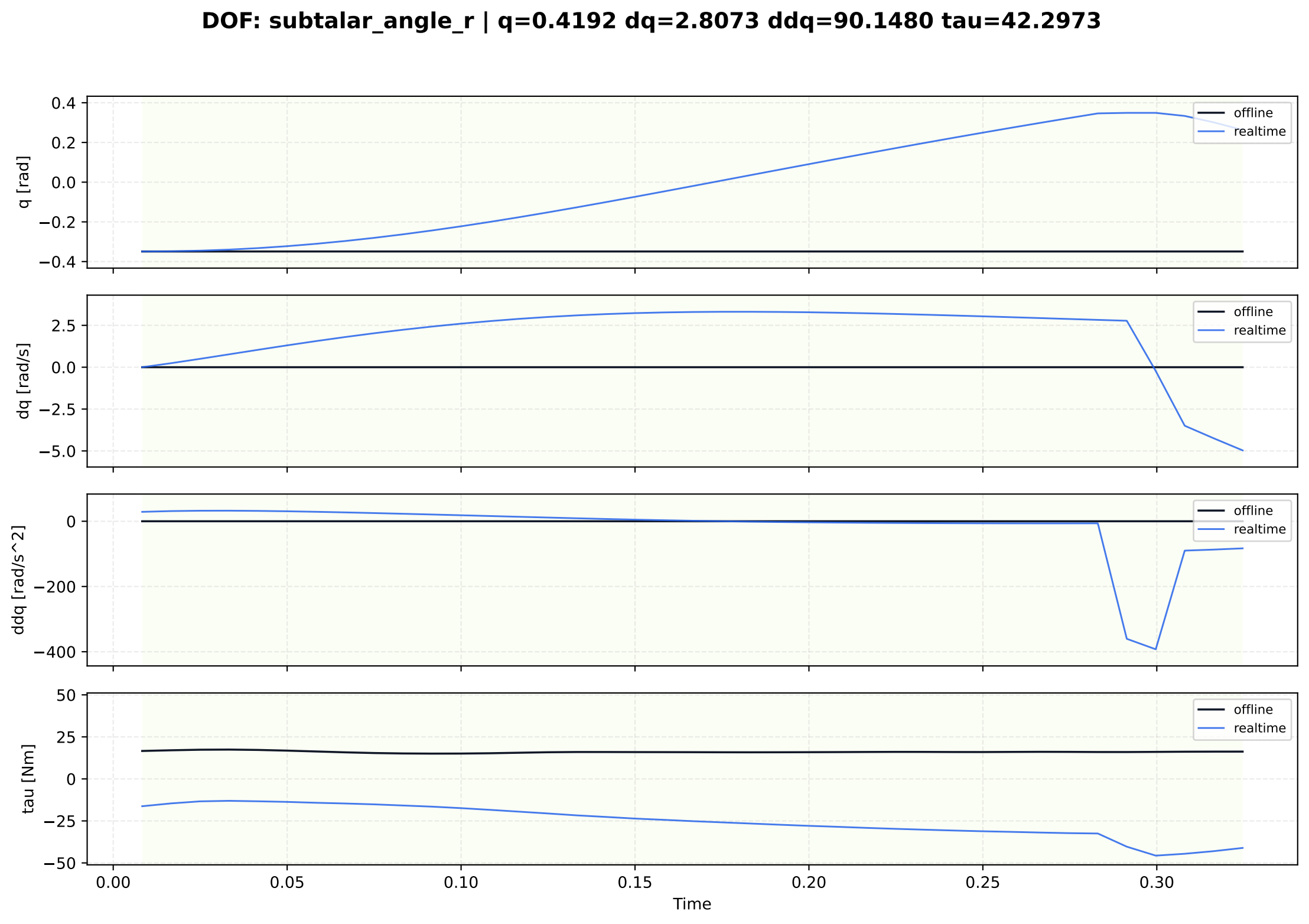

DOF: hip_flexion_r | q=0.0148 dq=0.3426 ddq=5.8854 tau=29.3944
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
Time
offline
realtime

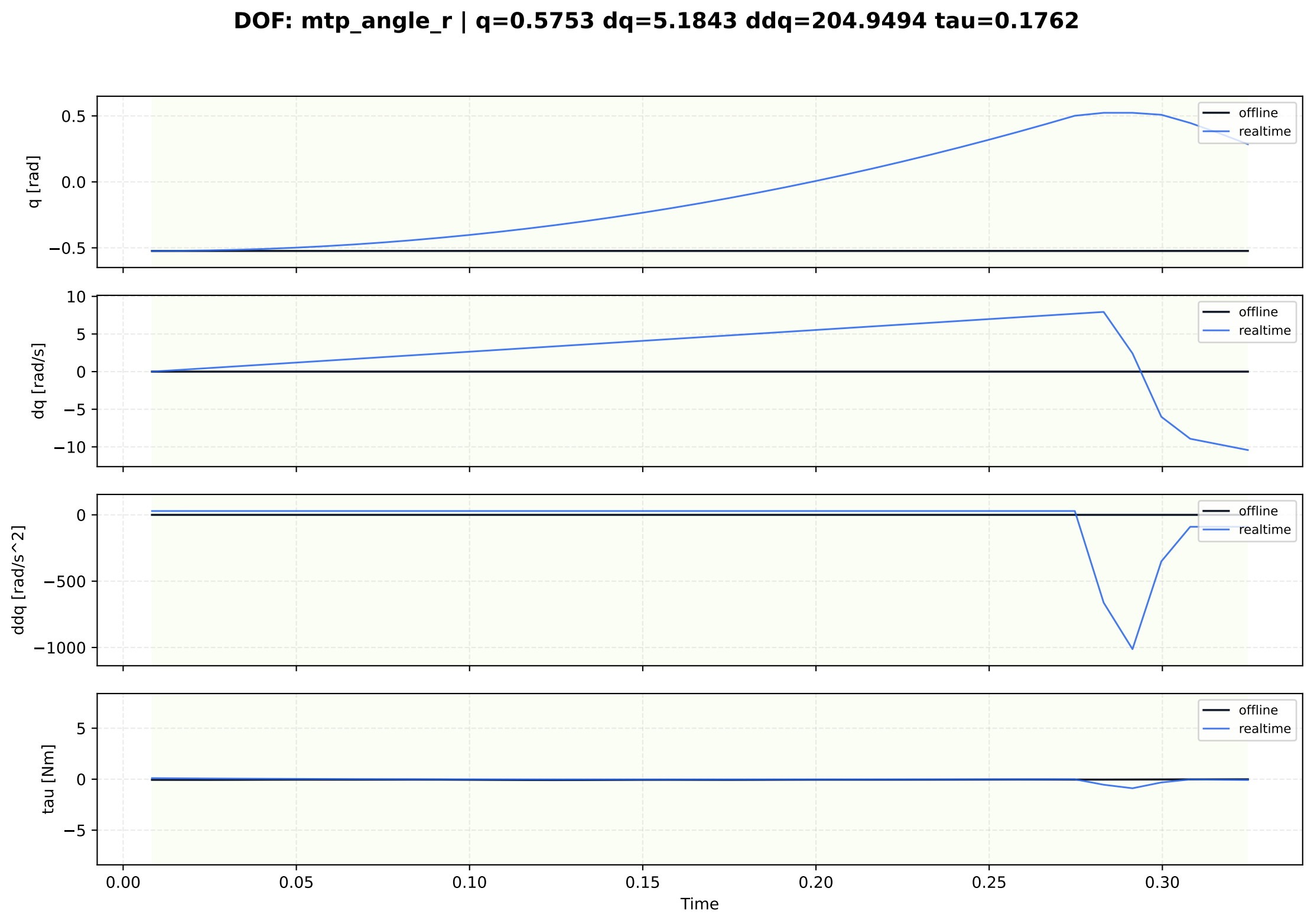

DOF: hip_adduction_r | q=0.0001 dq=1.0080 ddq=22.2160 tau=392.4661
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

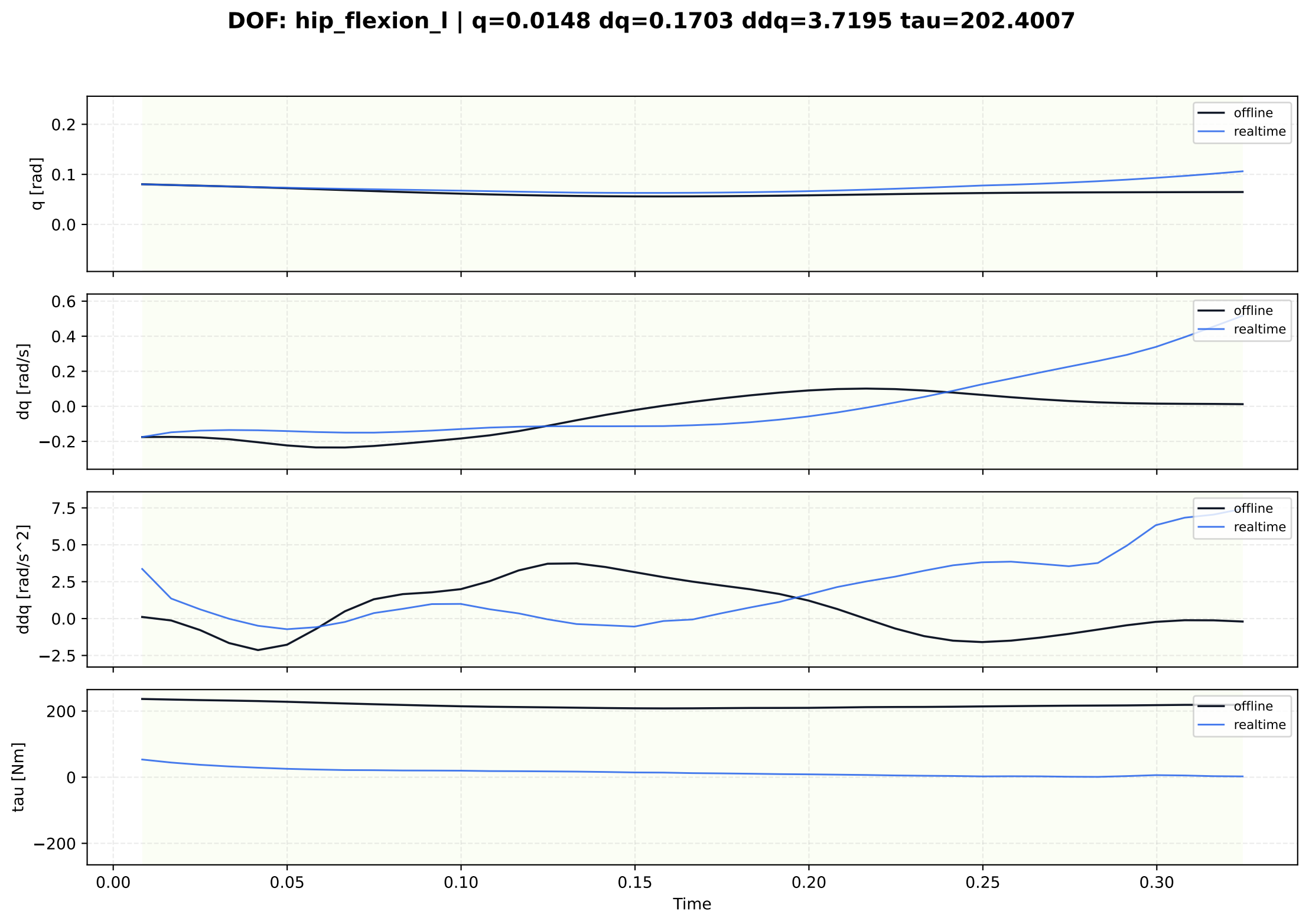

DOF: hip_rotation_r | q=0.0001 dq=0.4844 ddq=15.0842 tau=14.8343
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

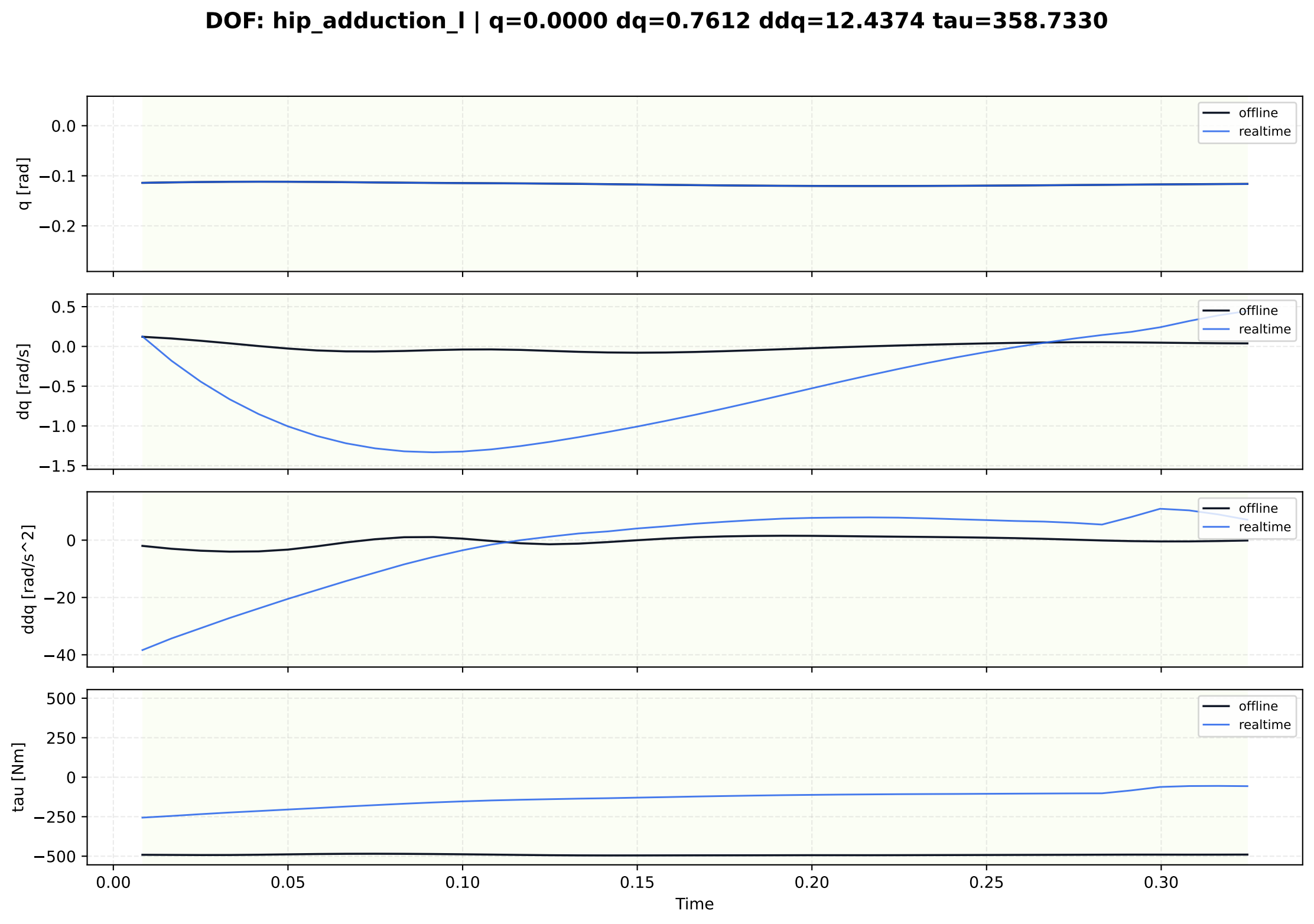

DOF: knee_angle_r | q=0.0000 dq=0.3720 ddq=8.7824 tau=350.0935
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

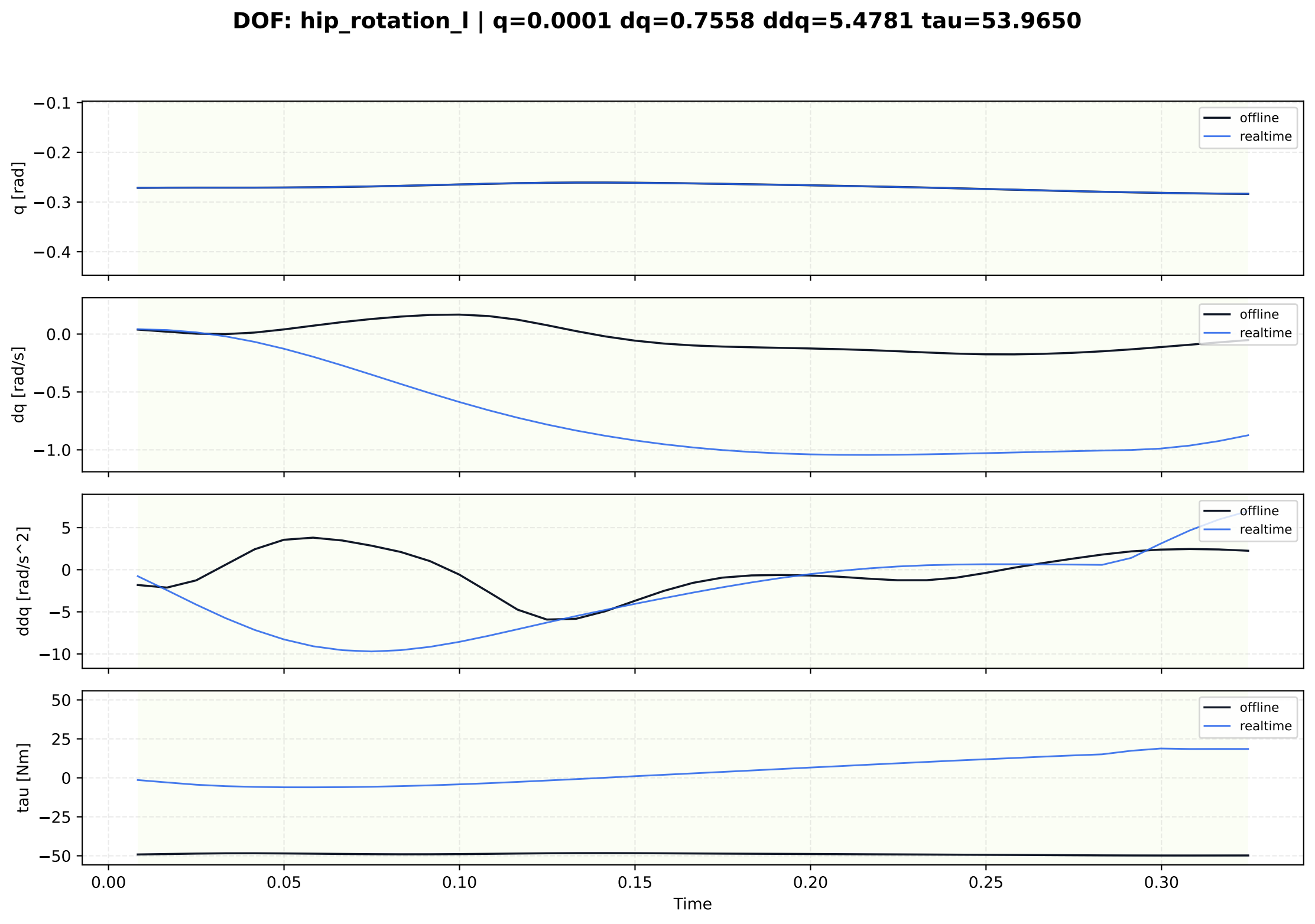

DOF: ankle_angle_r | q=0.0710 dq=0.3877 ddq=7.2479 tau=26.4255
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

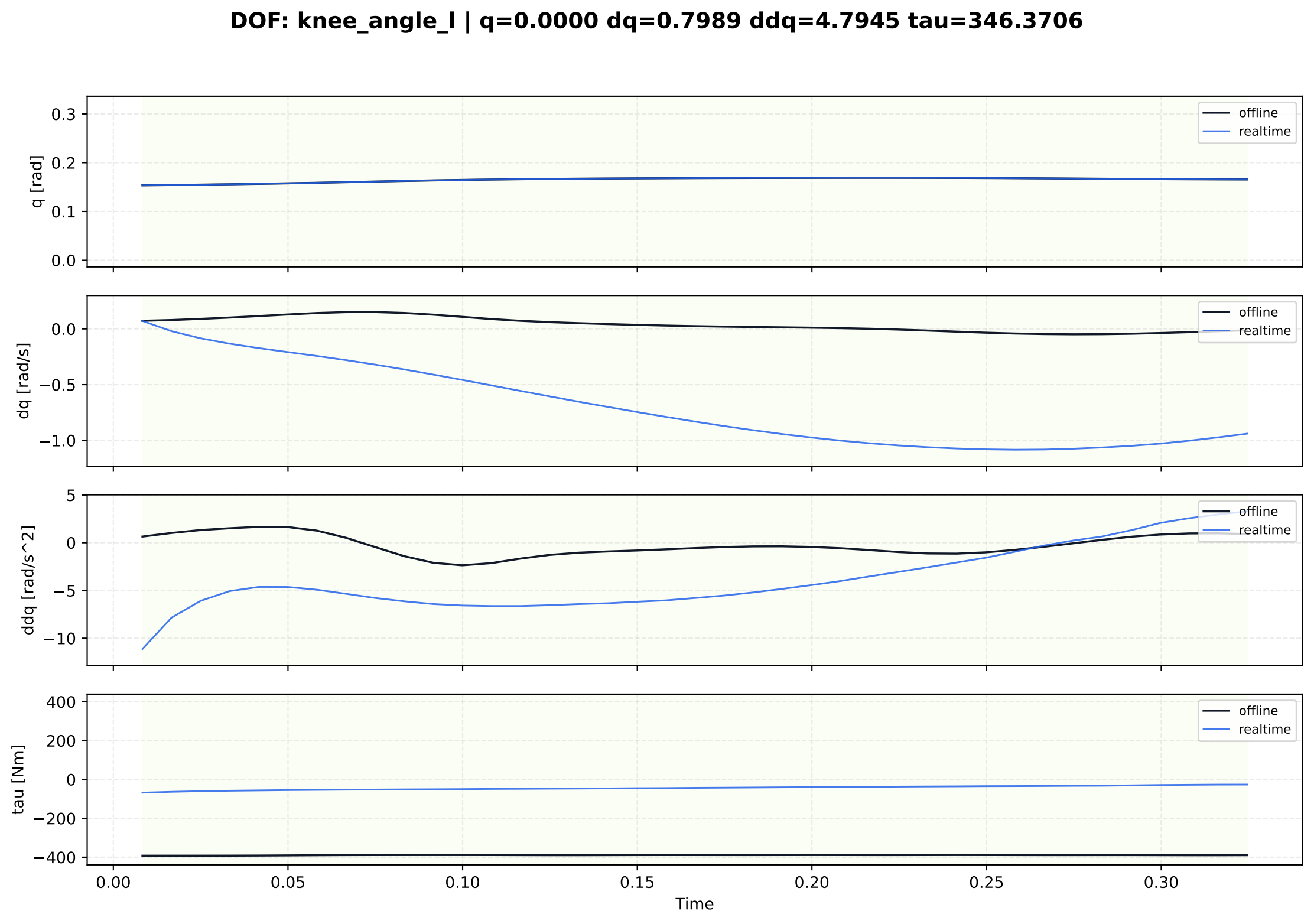

DOF: subtalar_angle_r | q=0.4192 dq=2.8073 ddq=90.1480 tau=42.2973
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

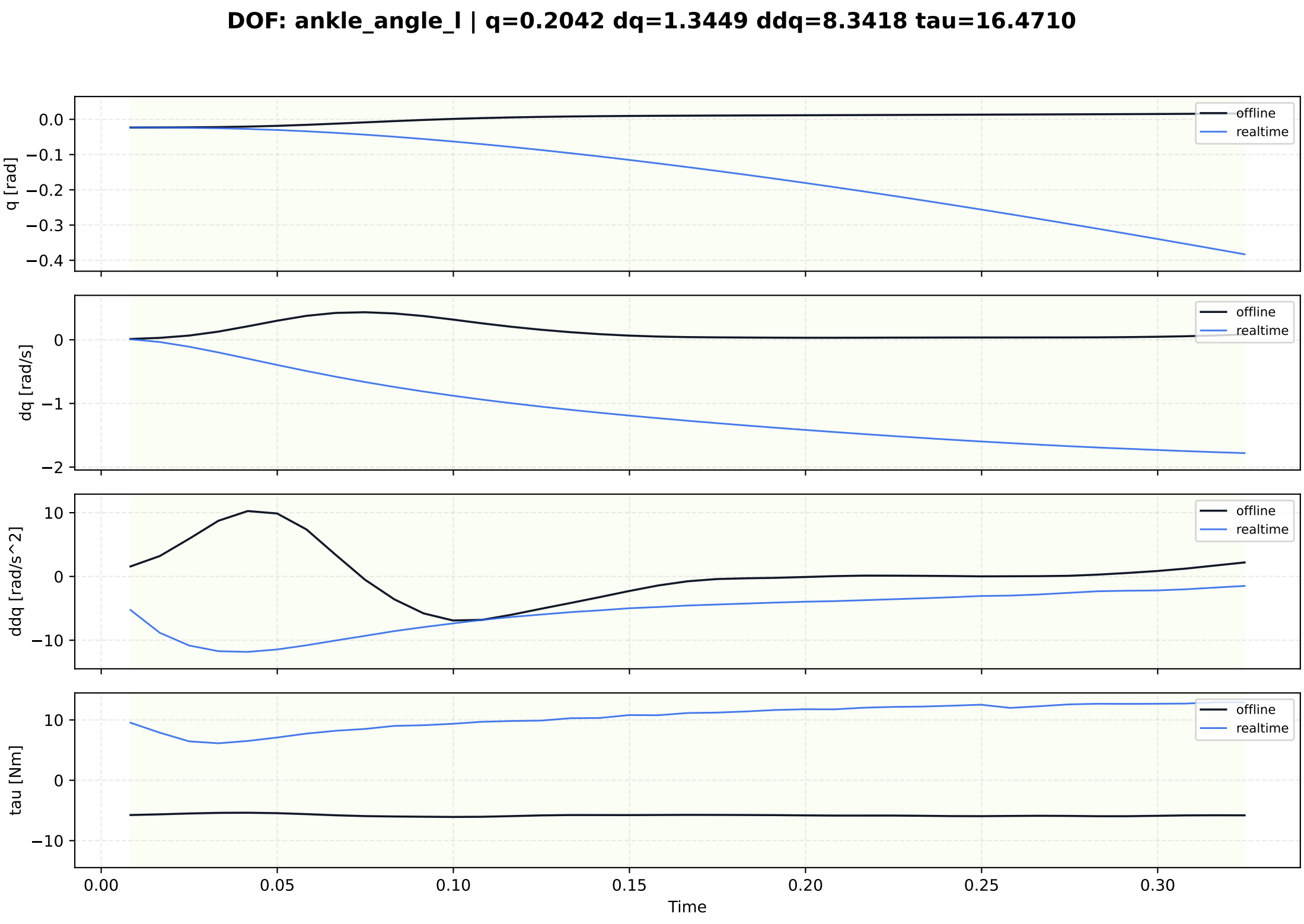

DOF: mtp_angle_r | q=0.5753 dq=5.1843 ddq=204.9494 tau=0.1762
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

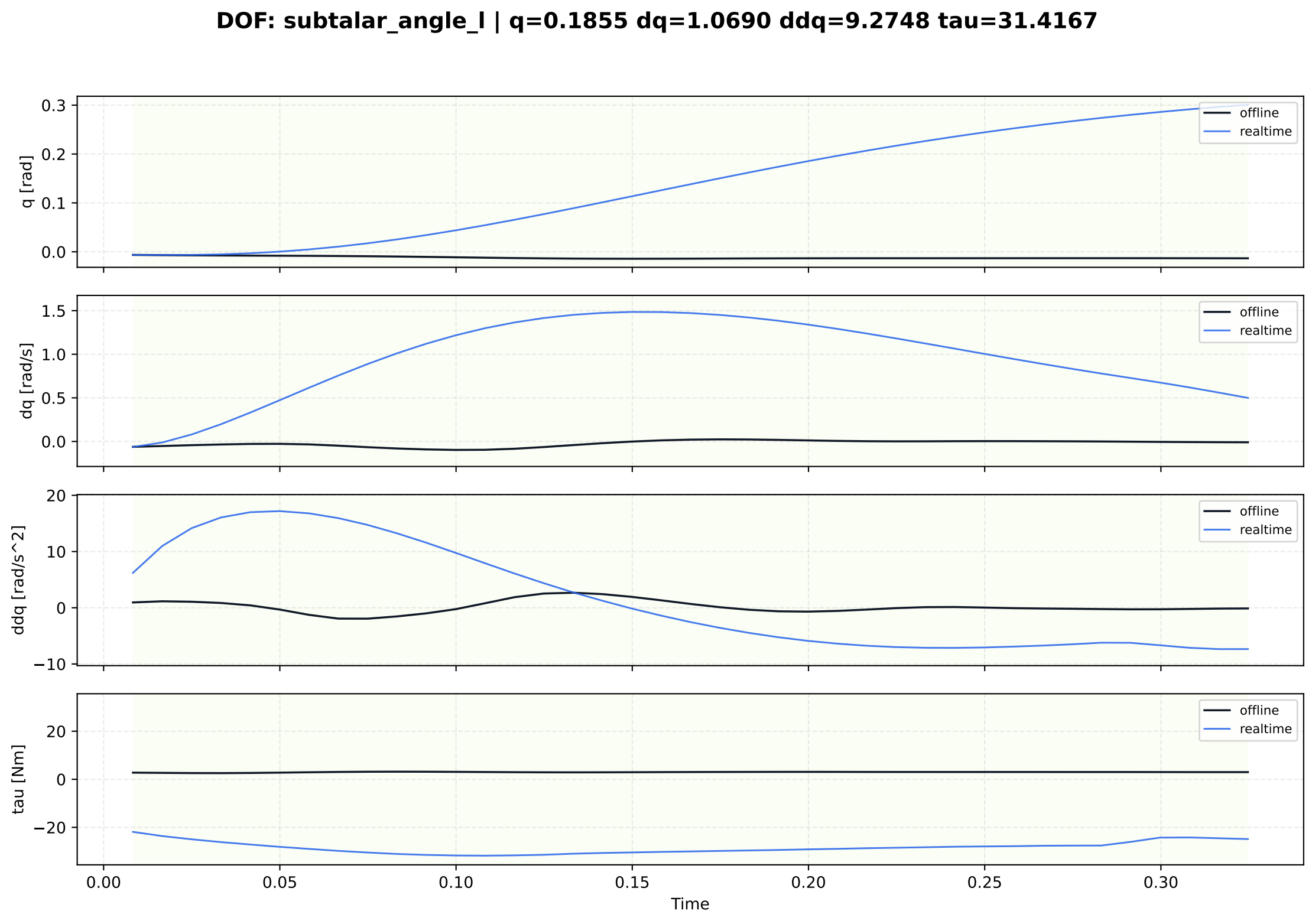

DOF: hip_flexion_l | q=0.0148 dq=0.1703 ddq=3.7195 tau=202.4007
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

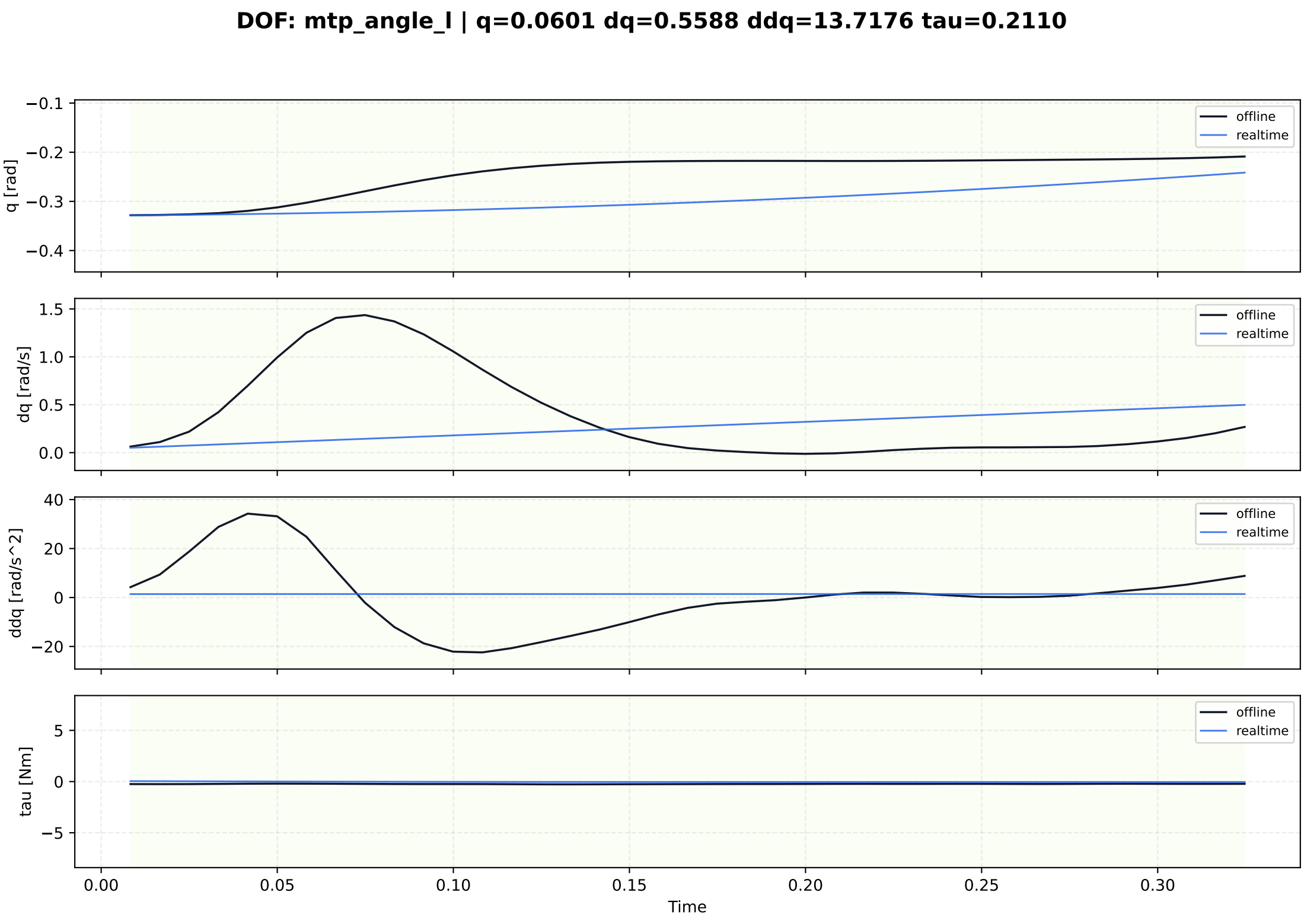

DOF: hip_adduction_l | q=0.0000 dq=0.7612 ddq=12.4374 tau=358.7330
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
Time
offline
realtime

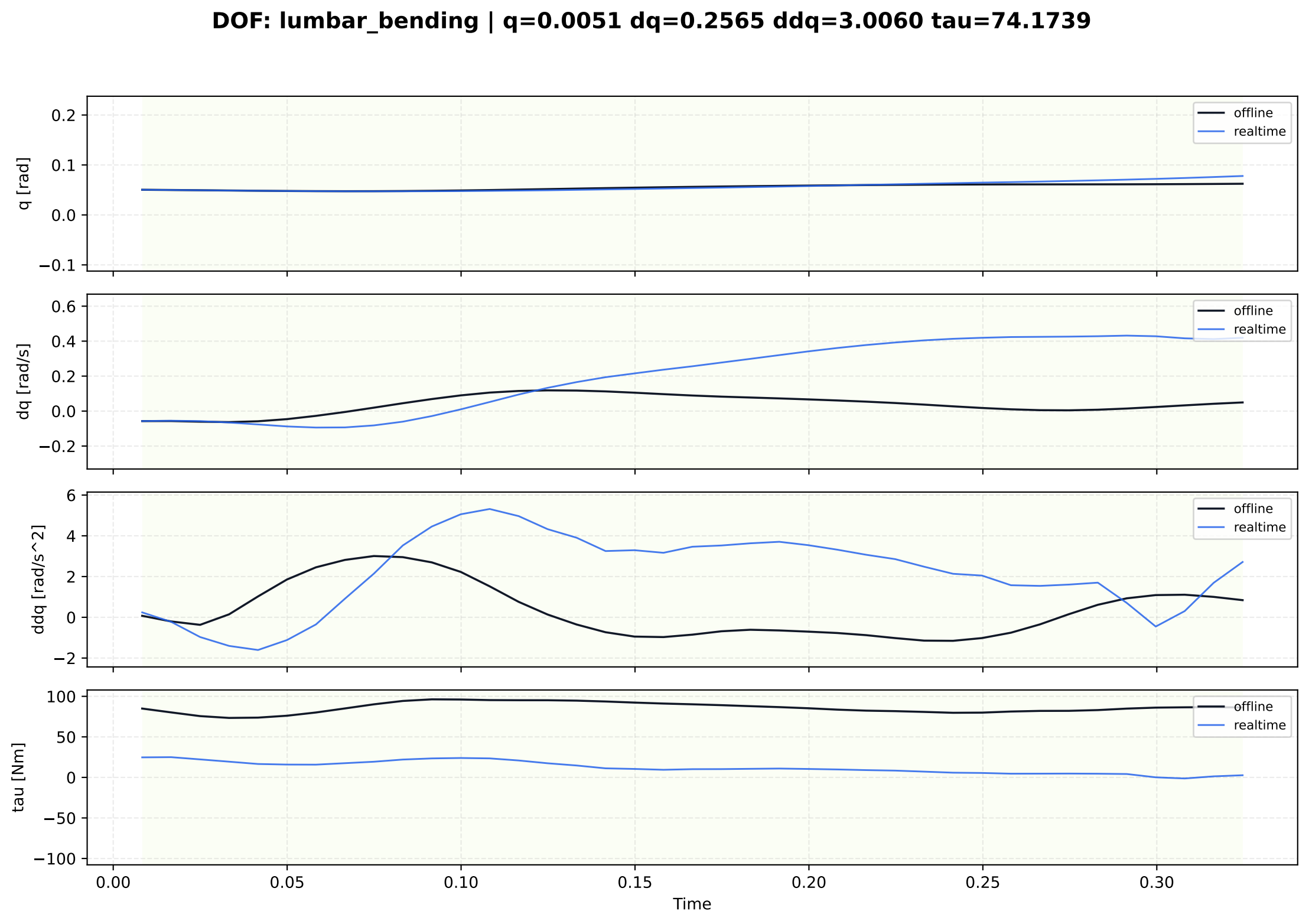

DOF: hip_rotation_l | q=0.0001 dq=0.7558 ddq=5.4781 tau=53.9650
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

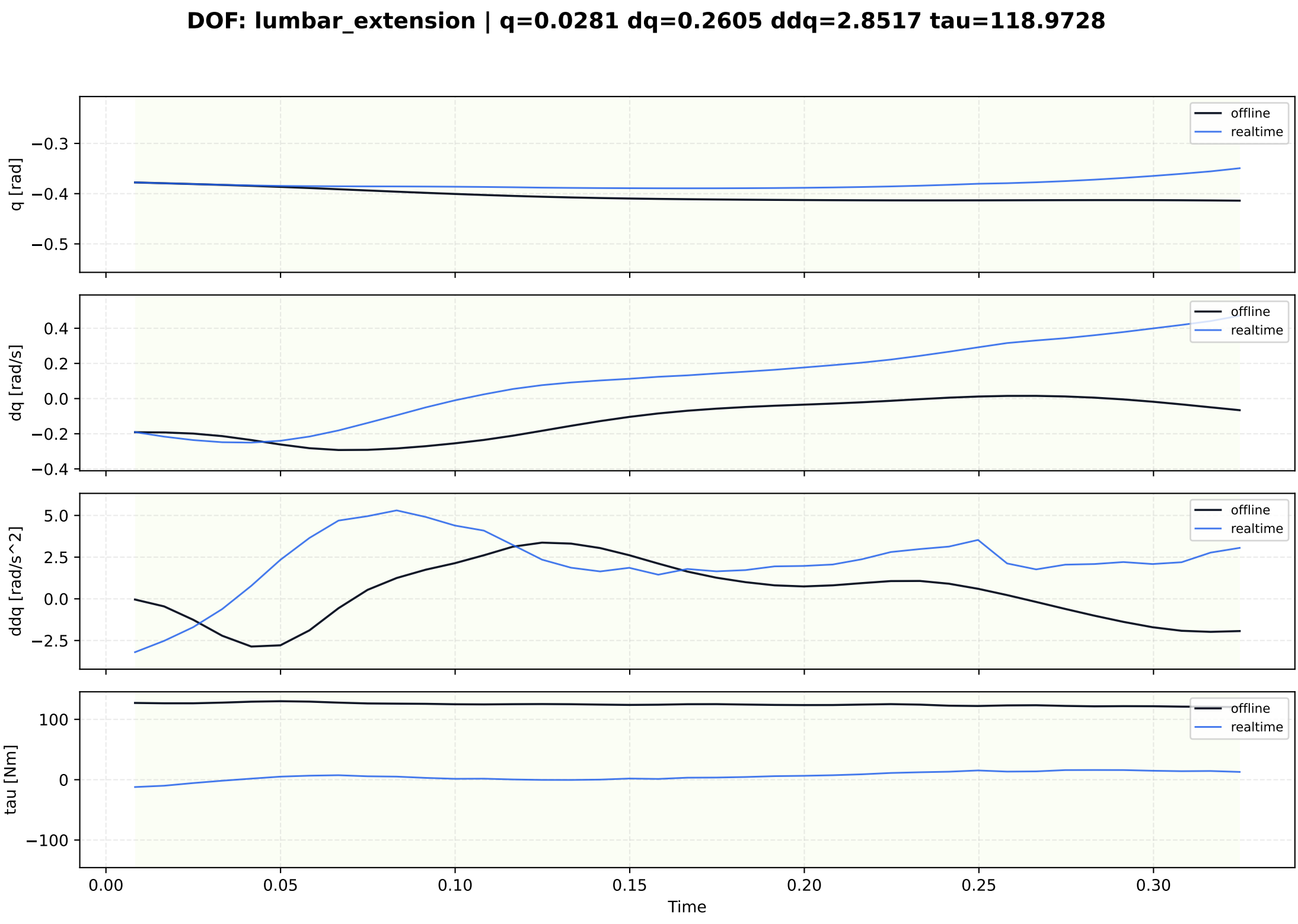

DOF: knee_angle_l | q=0.0000 dq=0.7989 ddq=4.7945 tau=346.3706
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

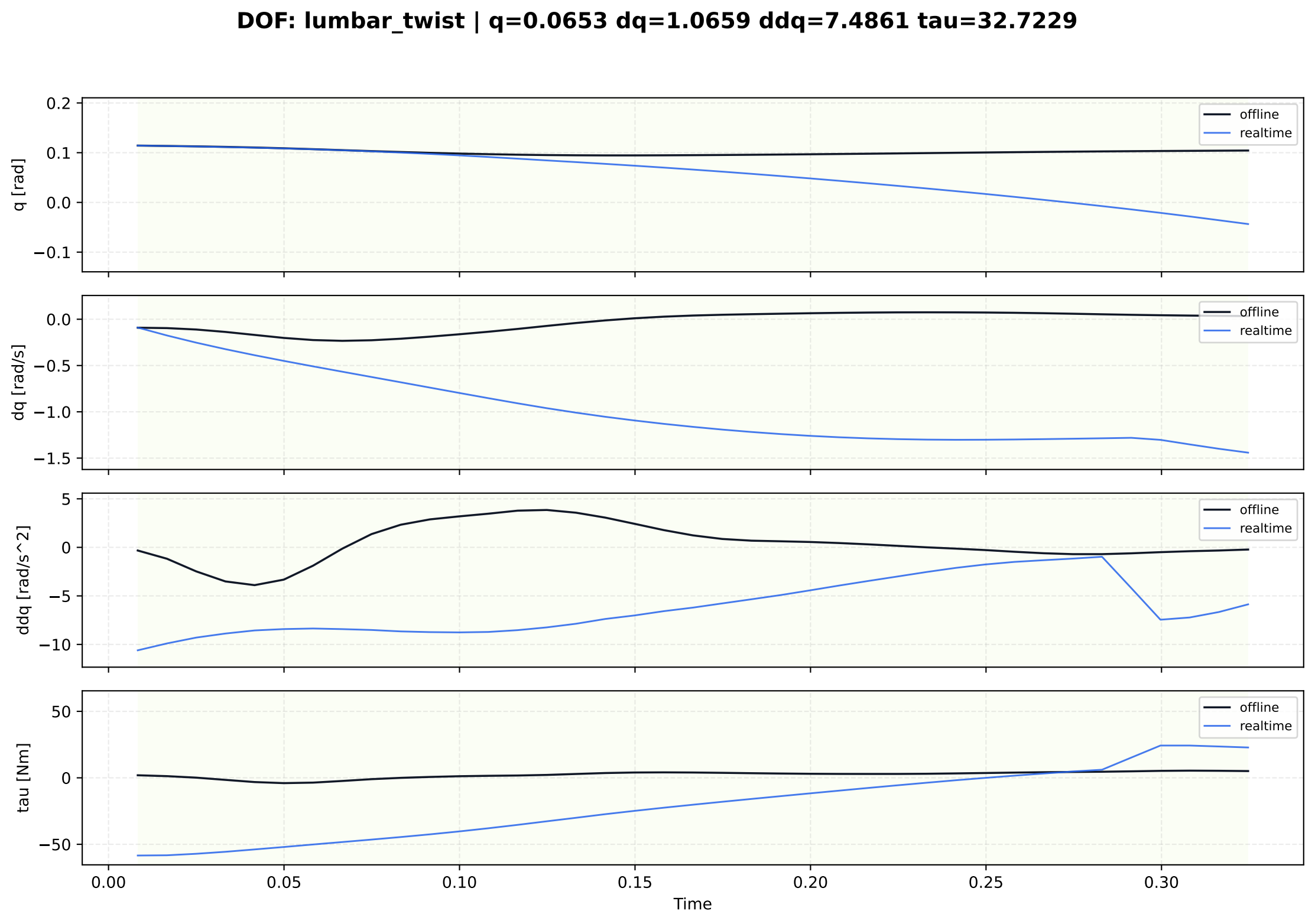

DOF: ankle_angle_l | q=0.2042 dq=1.3449 ddq=8.3418 tau=16.4710
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

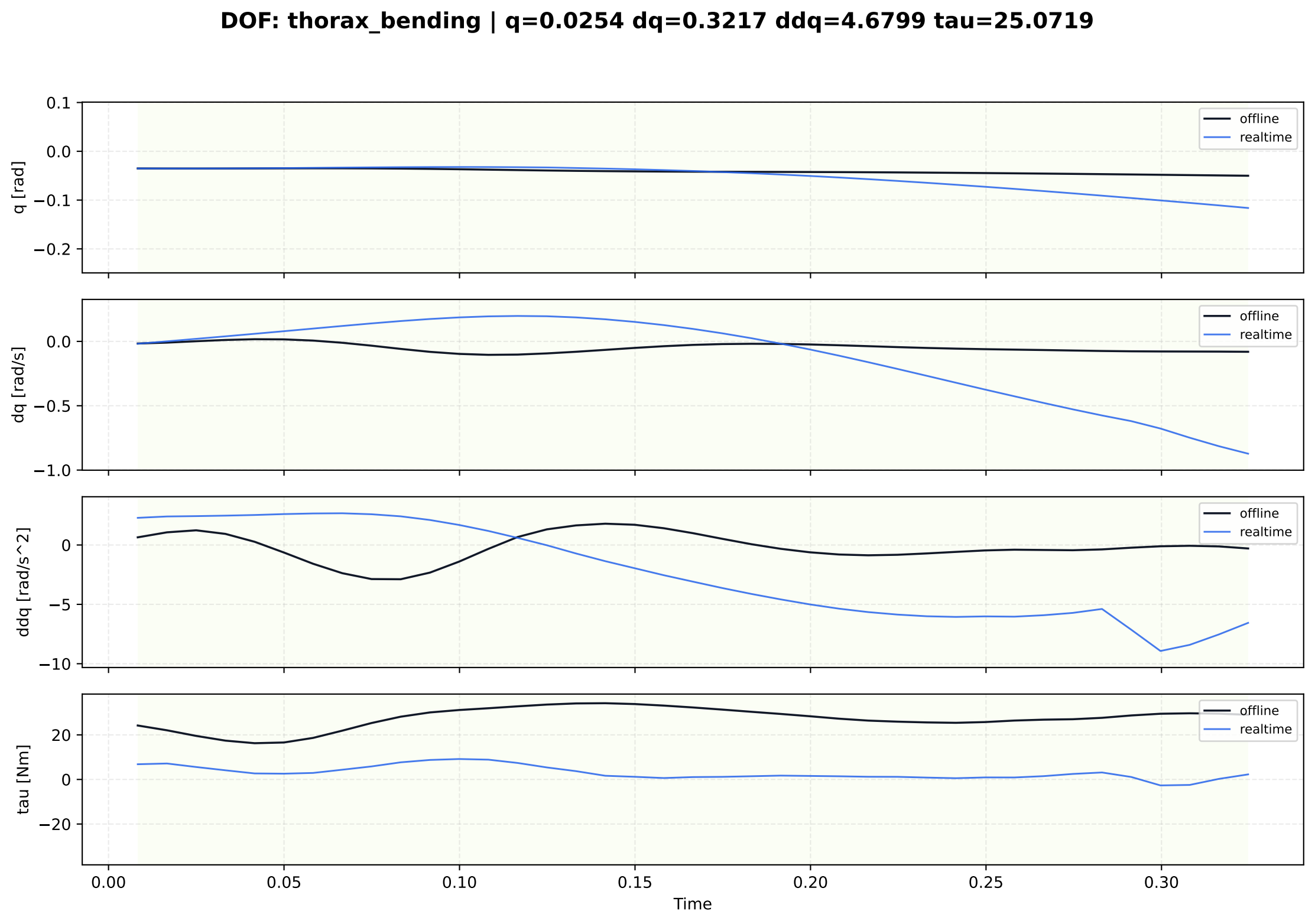

DOF: subtalar_angle_l | q=0.1855 dq=1.0690 ddq=9.2748 tau=31.4167
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

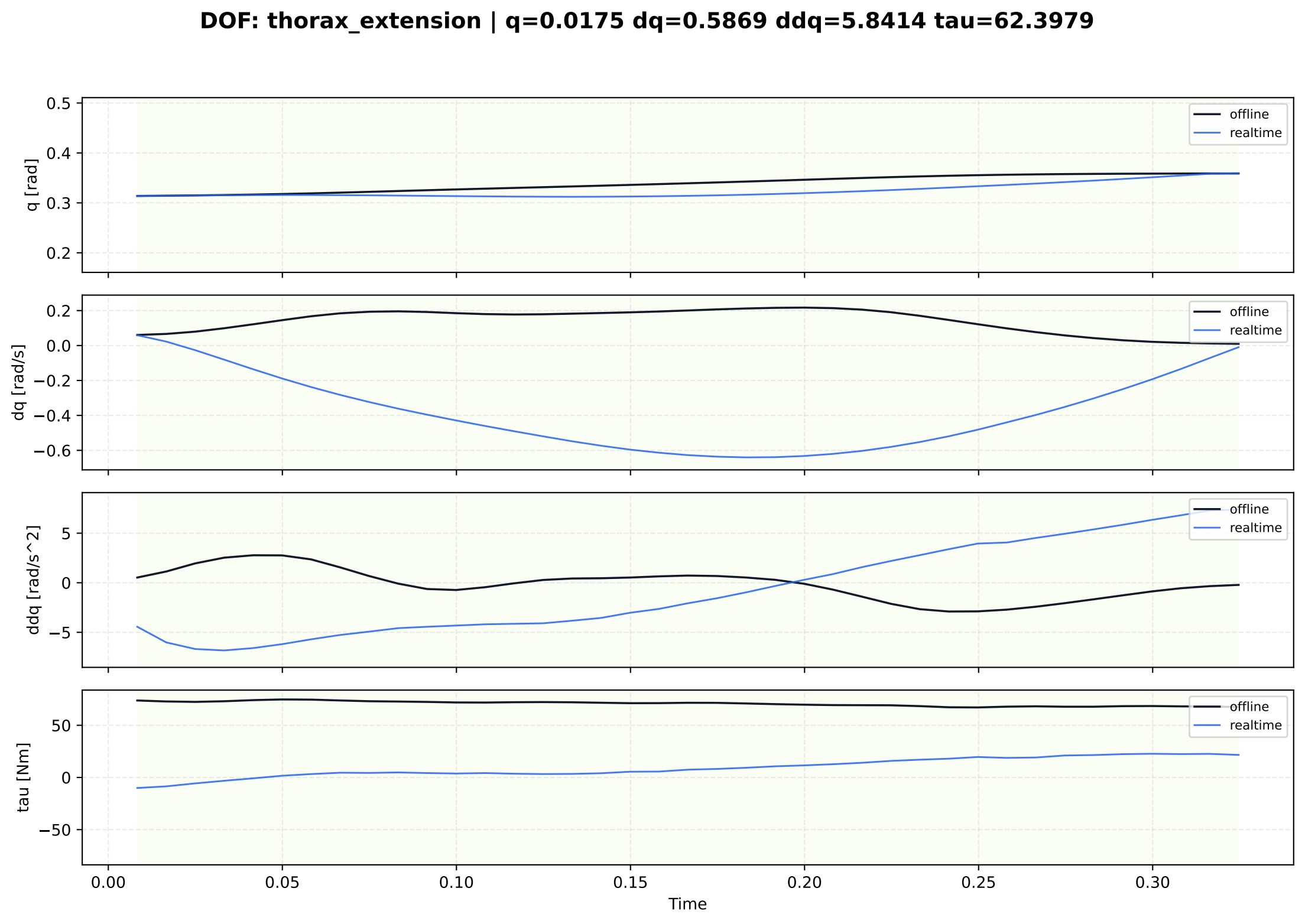

DOF: mtp_angle_l | q=0.0601 dq=0.5588 ddq=13.7176 tau=0.2110
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

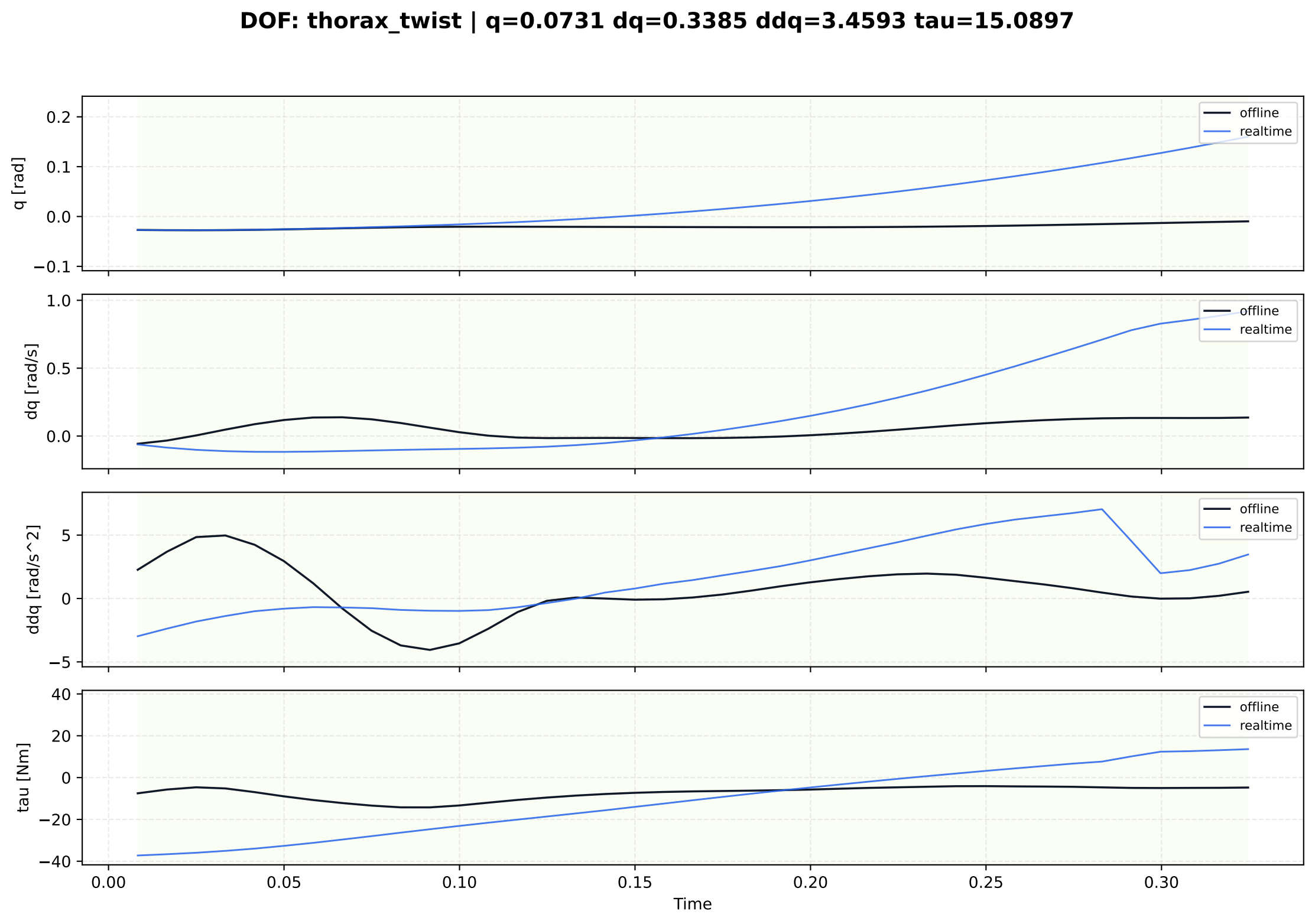

DOF: lumbar_bending | q=0.0051 dq=0.2565 ddq=3.0060 tau=74.1739
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

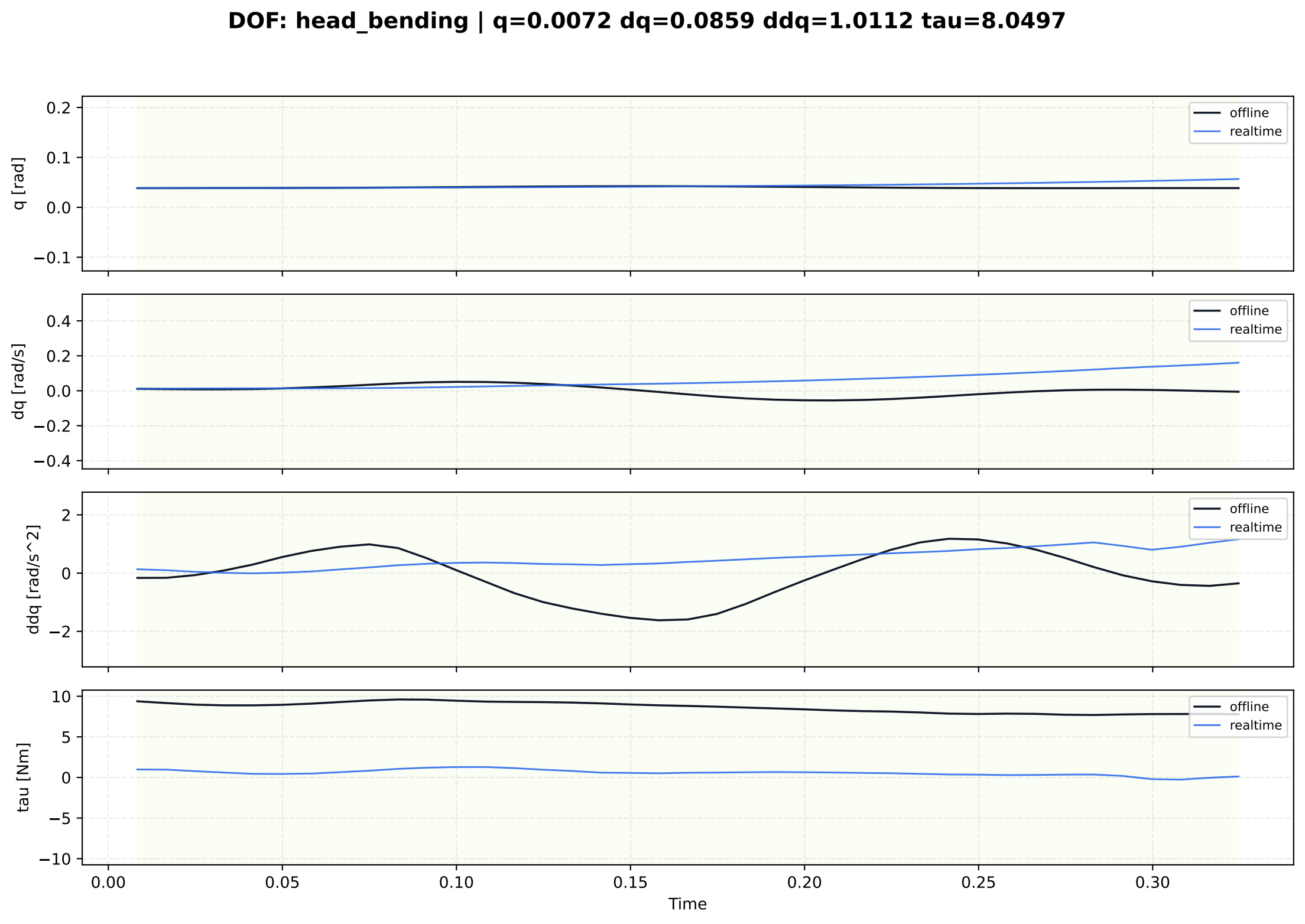

DOF: lumbar_extension | q=0.0281 dq=0.2605 ddq=2.8517 tau=118.9728
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

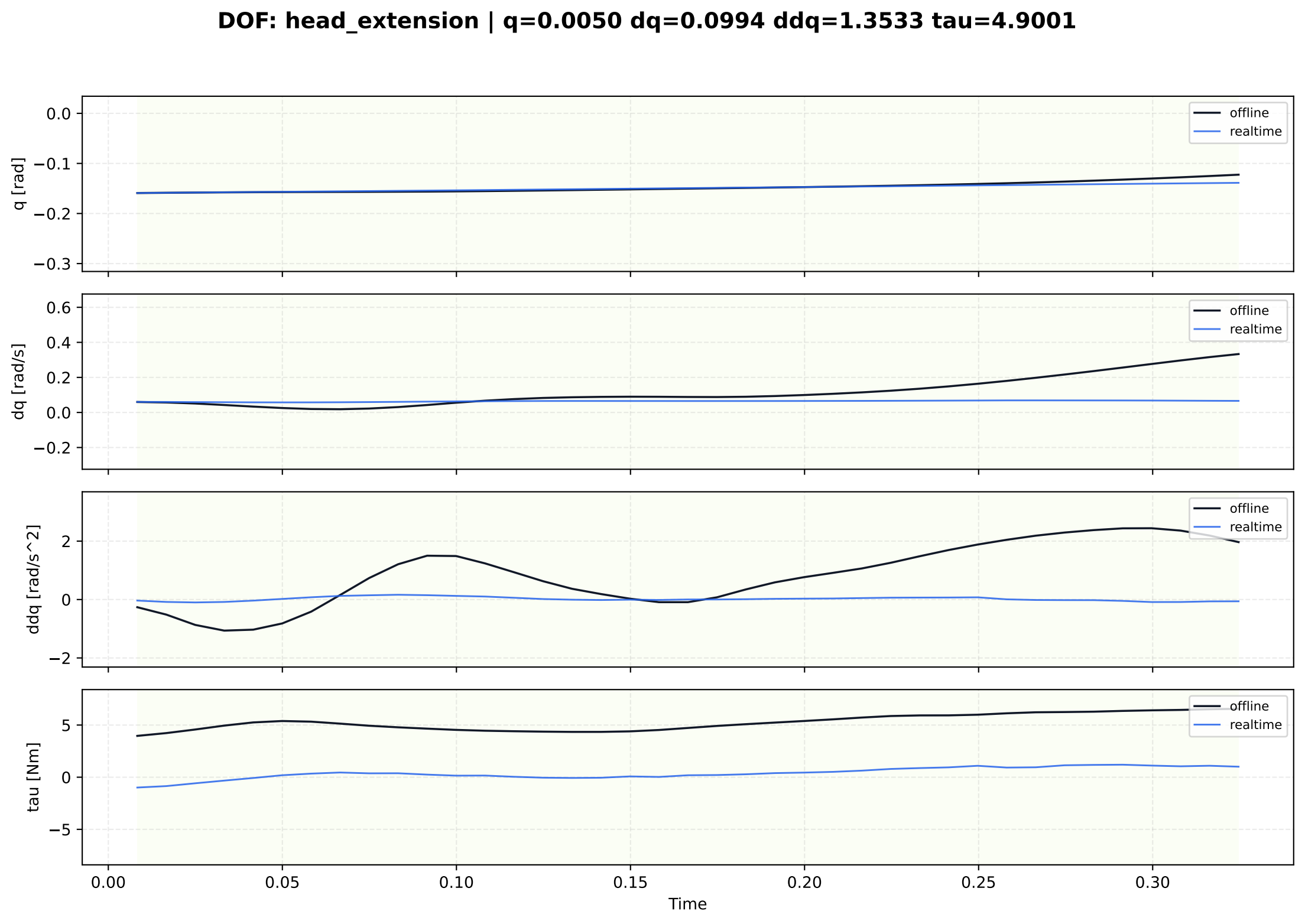

DOF: lumbar_twist | q=0.0653 dq=1.0659 ddq=7.4861 tau=32.7229
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

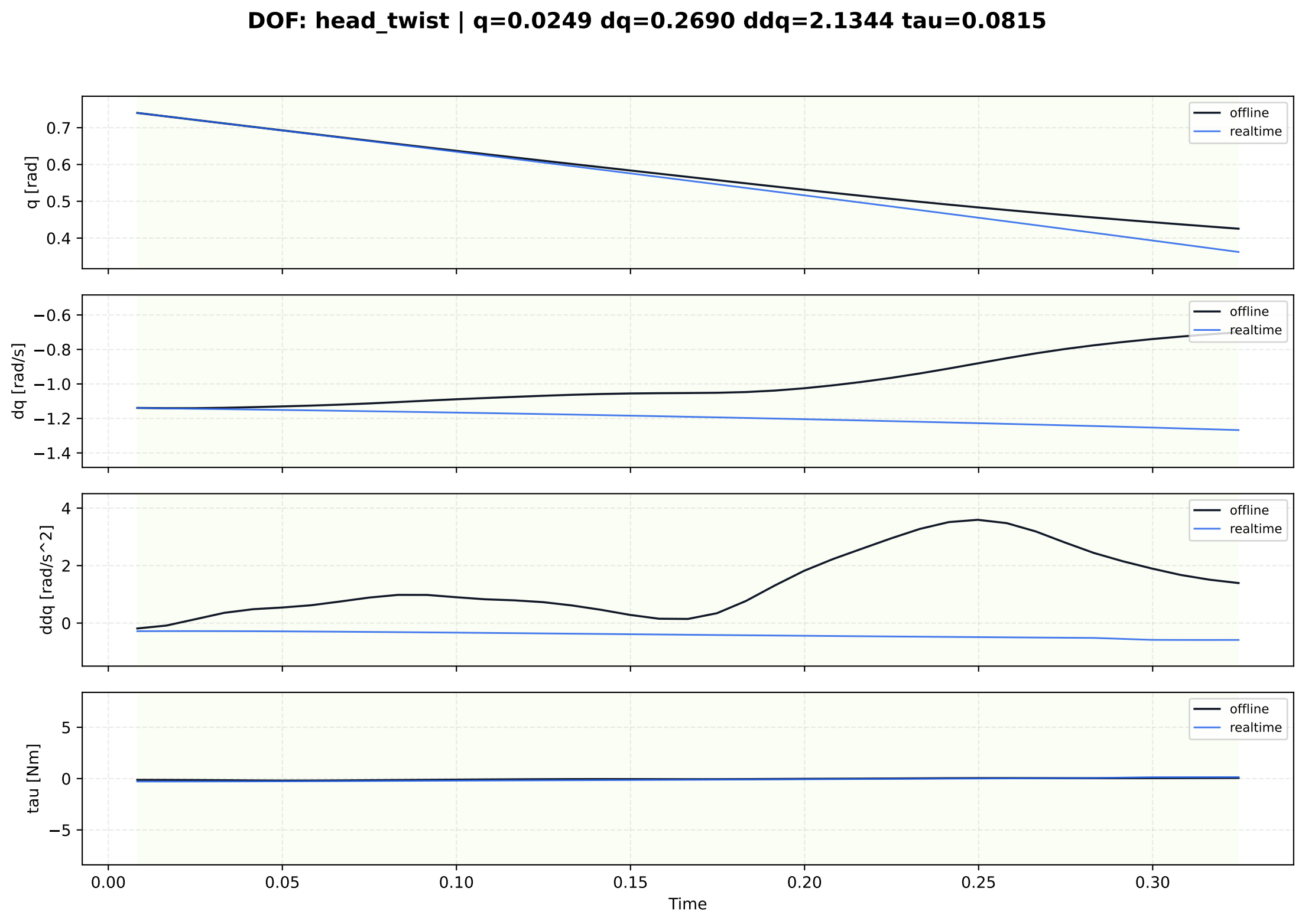

DOF: thorax_bending | q=0.0254 dq=0.3217 ddq=4.6799 tau=25.0719
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

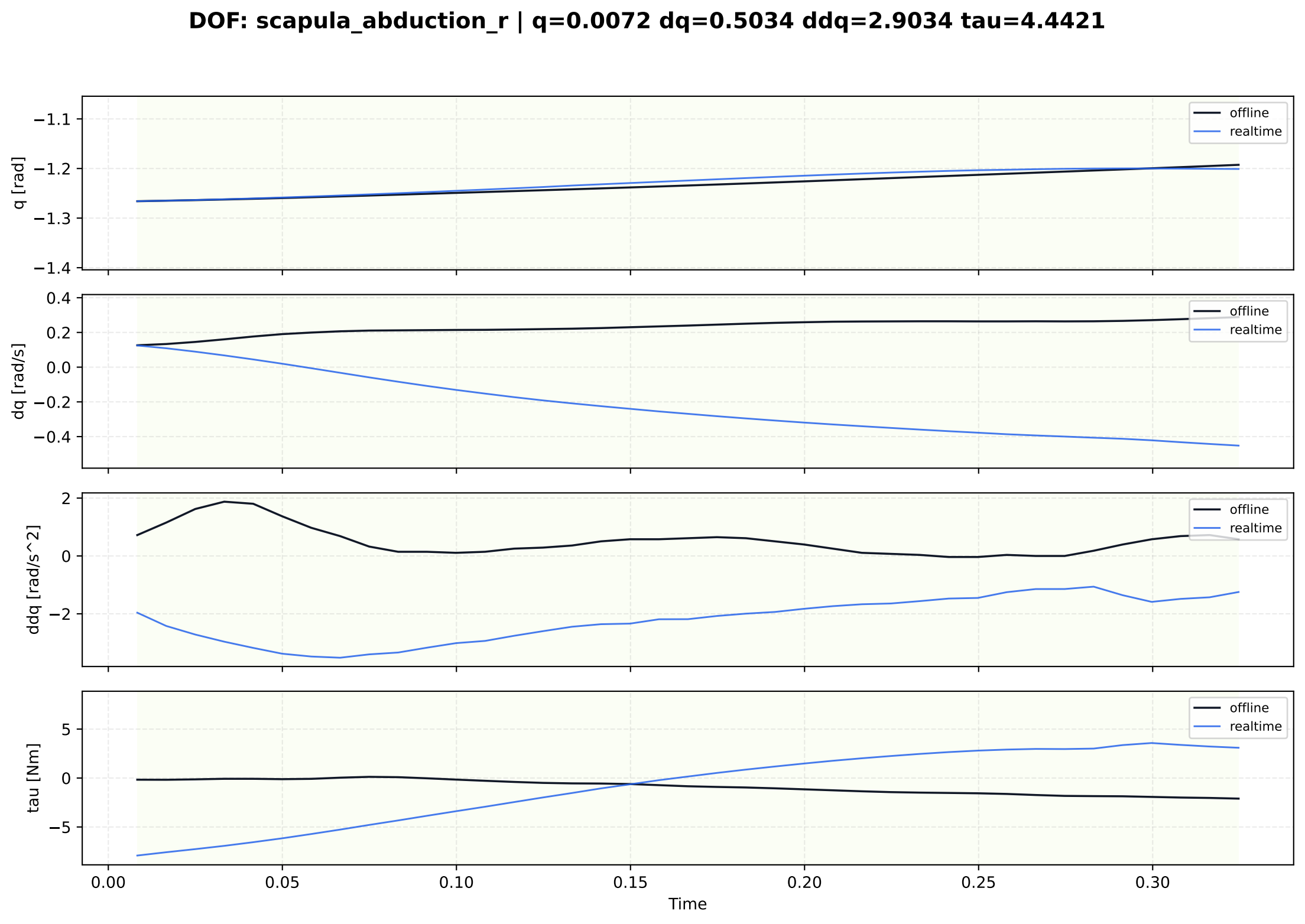

DOF: thorax_extension | q=0.0175 dq=0.5869 ddq=5.8414 tau=62.3979
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

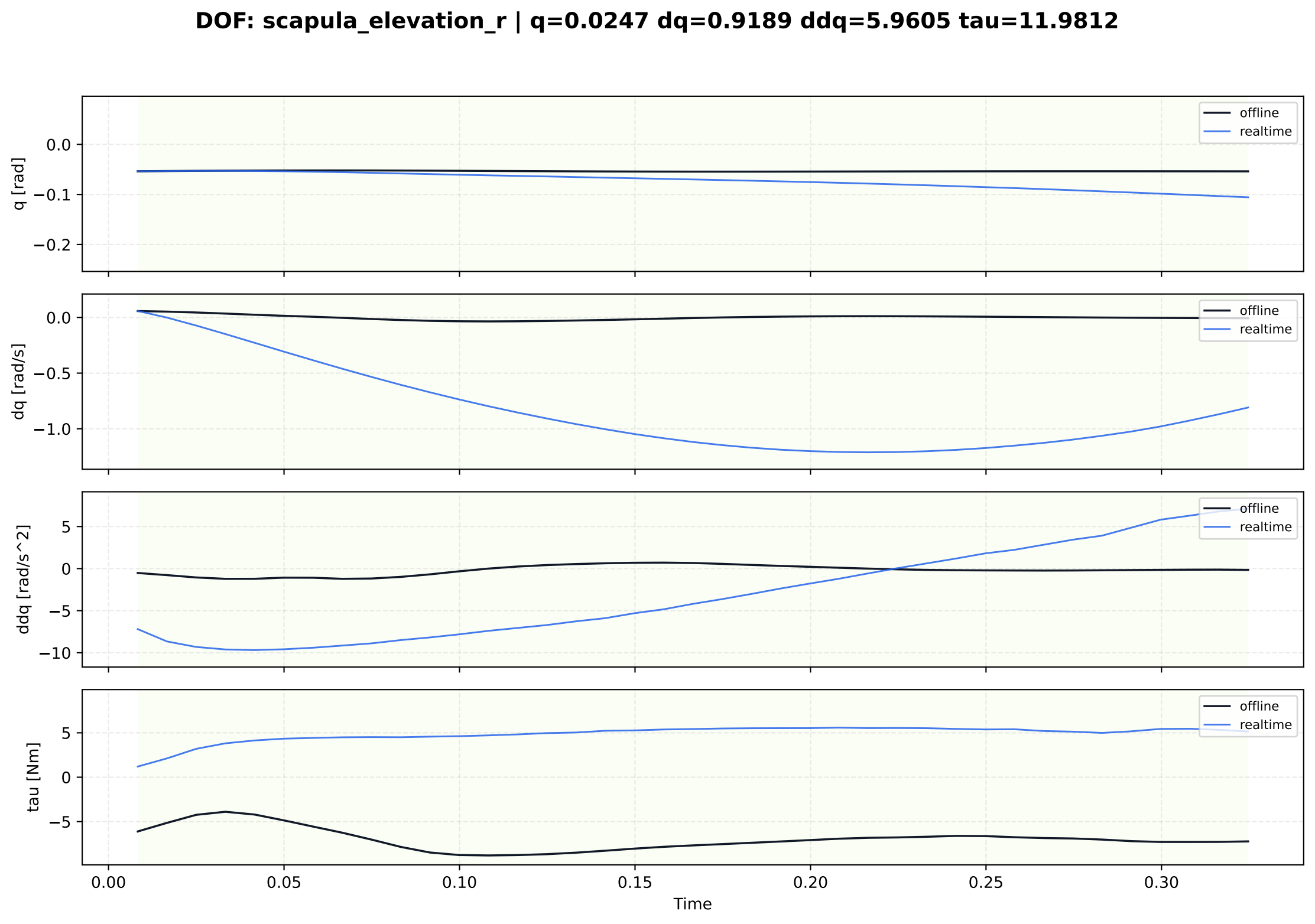

DOF: thorax_twist | q=0.0731 dq=0.3385 ddq=3.4593 tau=15.0897
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

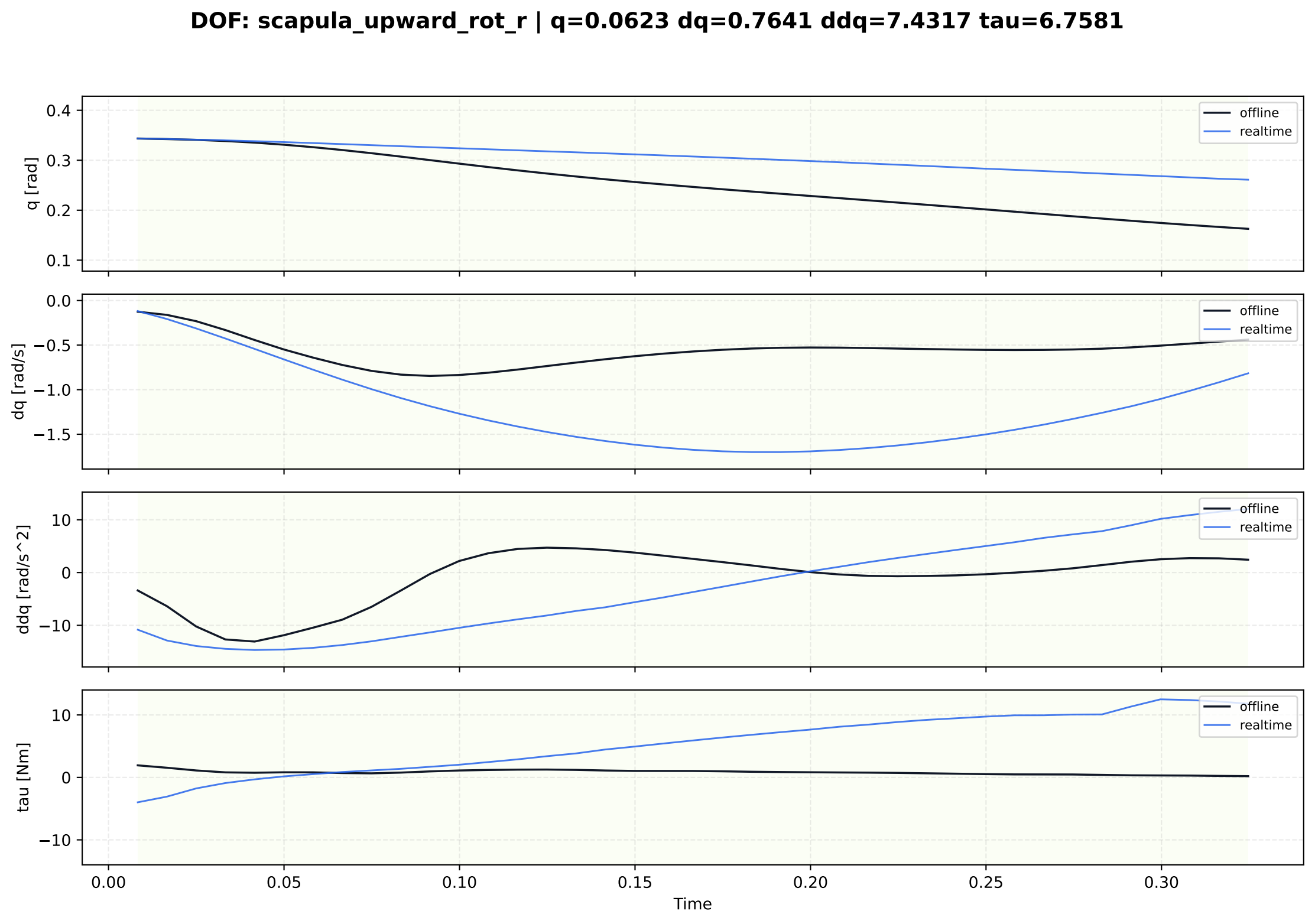

DOF: head_bending | q=0.0072 dq=0.0859 ddq=1.0112 tau=8.0497
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

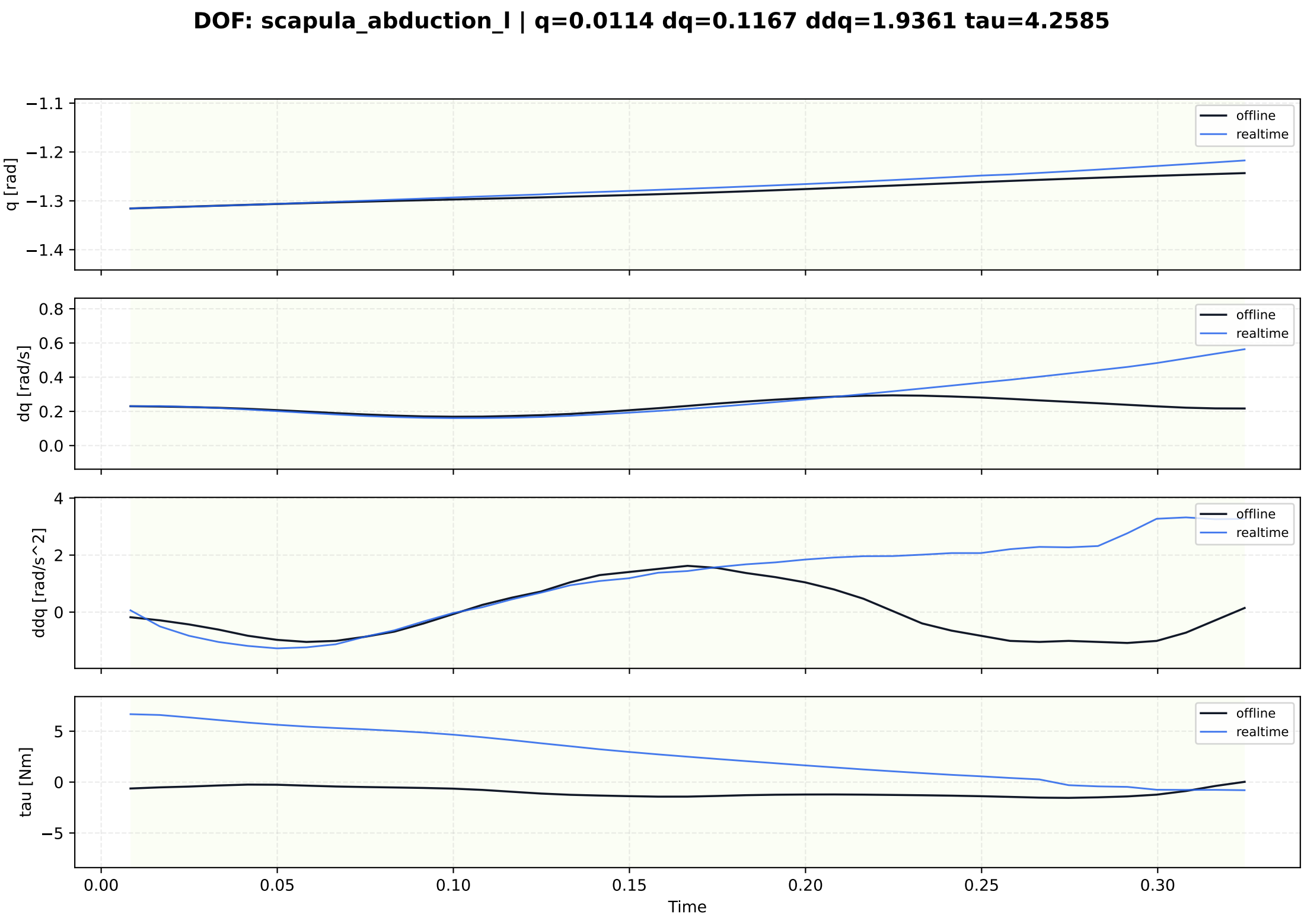

DOF: head_extension | q=0.0050 dq=0.0994 ddq=1.3533 tau=4.9001
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

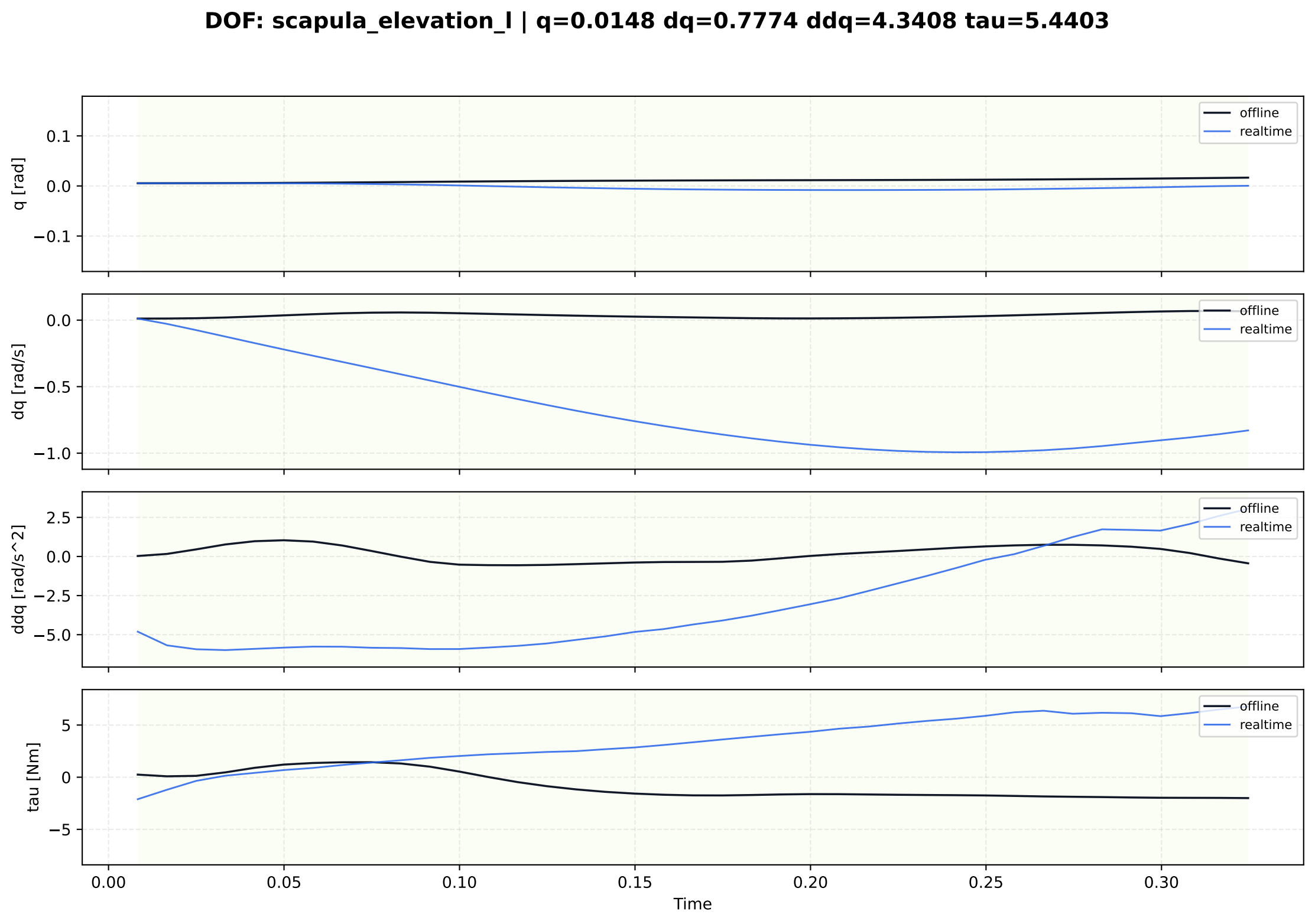

DOF: head_twist | q=0.0249 dq=0.2690 ddq=2.1344 tau=0.0815
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

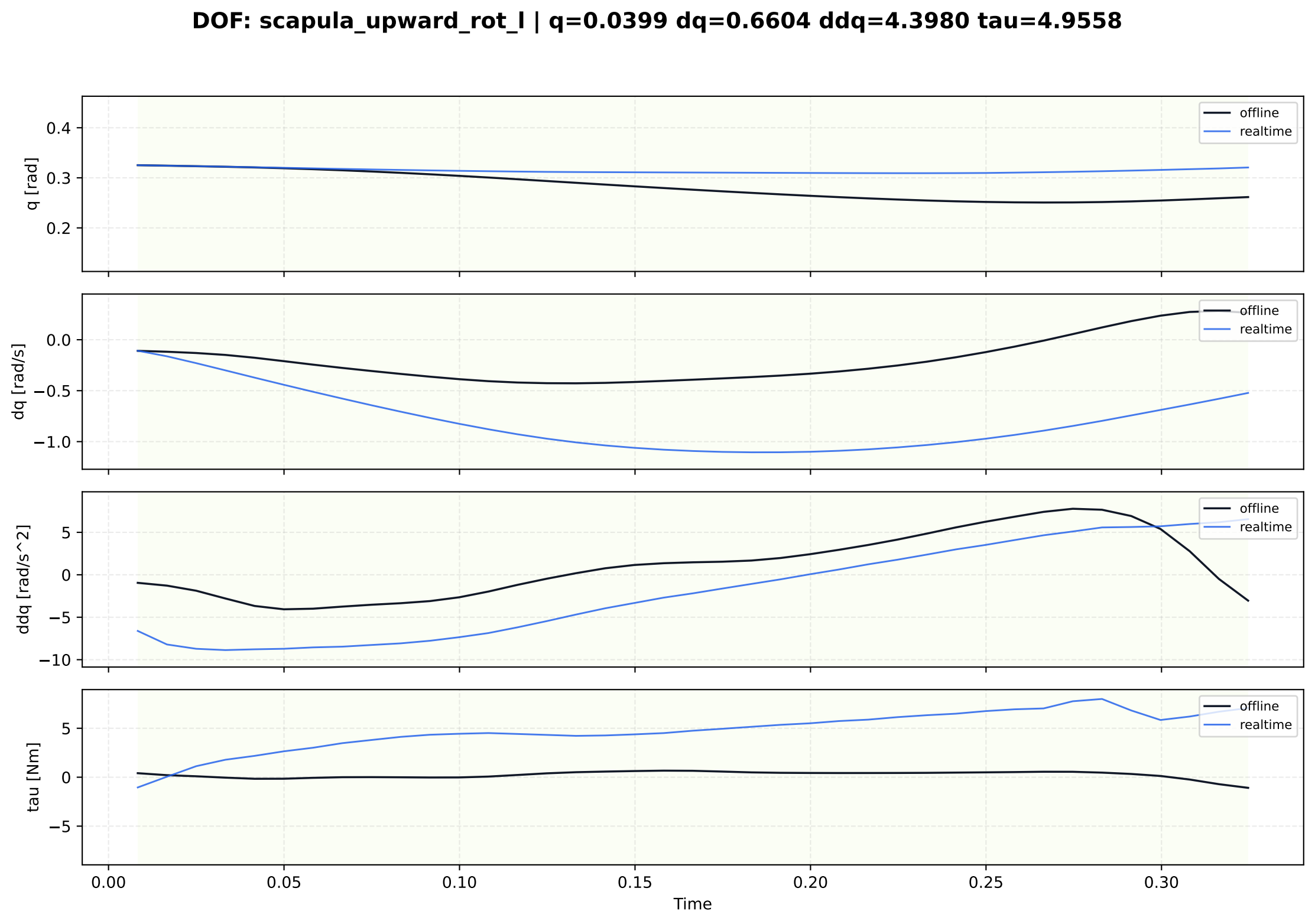

DOF: scapula_abduction_r | q=0.0072 dq=0.5034 ddq=2.9034 tau=4.4421
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

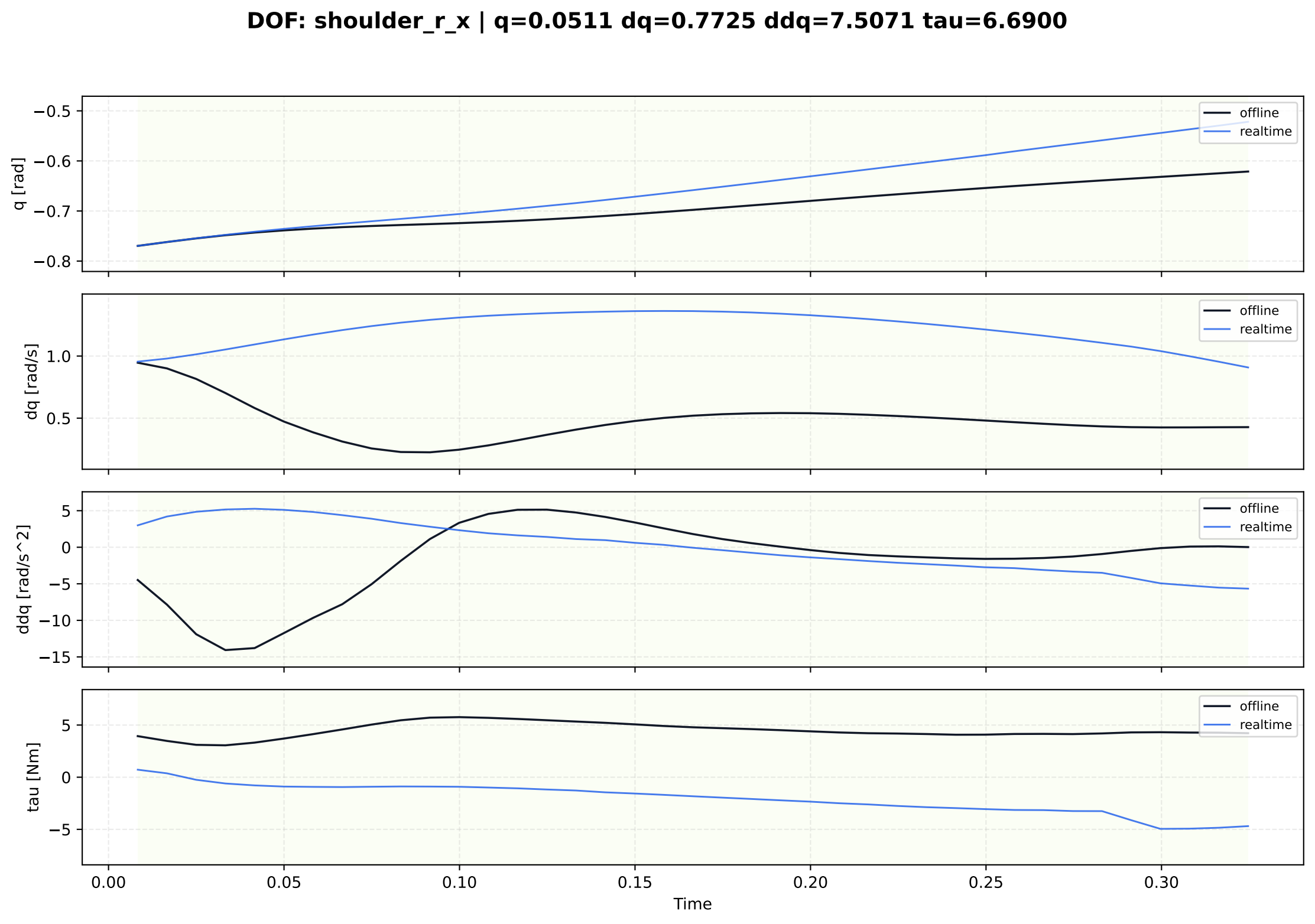

DOF: scapula_elevation_r | q=0.0247 dq=0.9189 ddq=5.9605 tau=11.9812
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

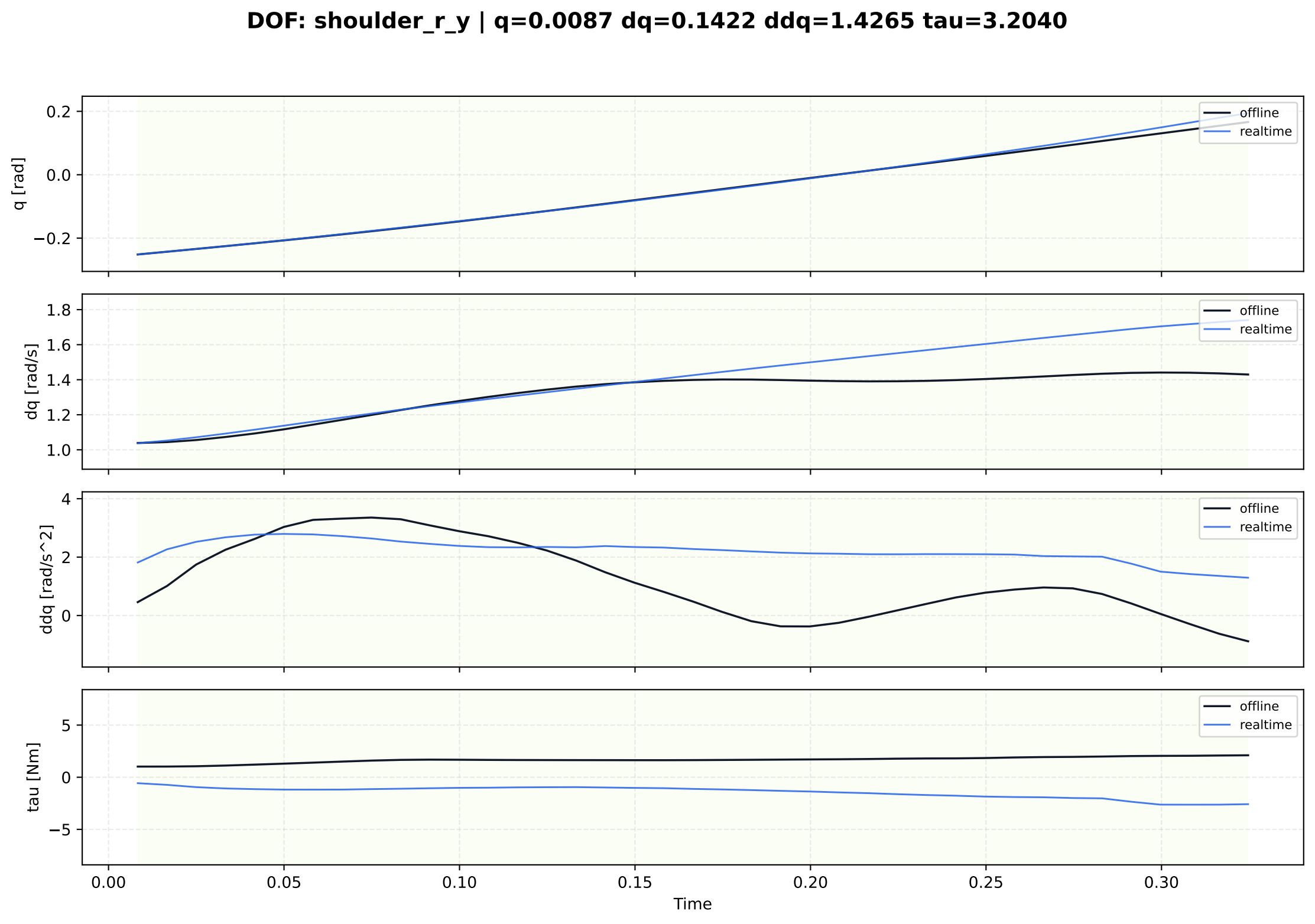

DOF: scapula_upward_rot_r | q=0.0623 dq=0.7641 ddq=7.4317 tau=6.7581
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

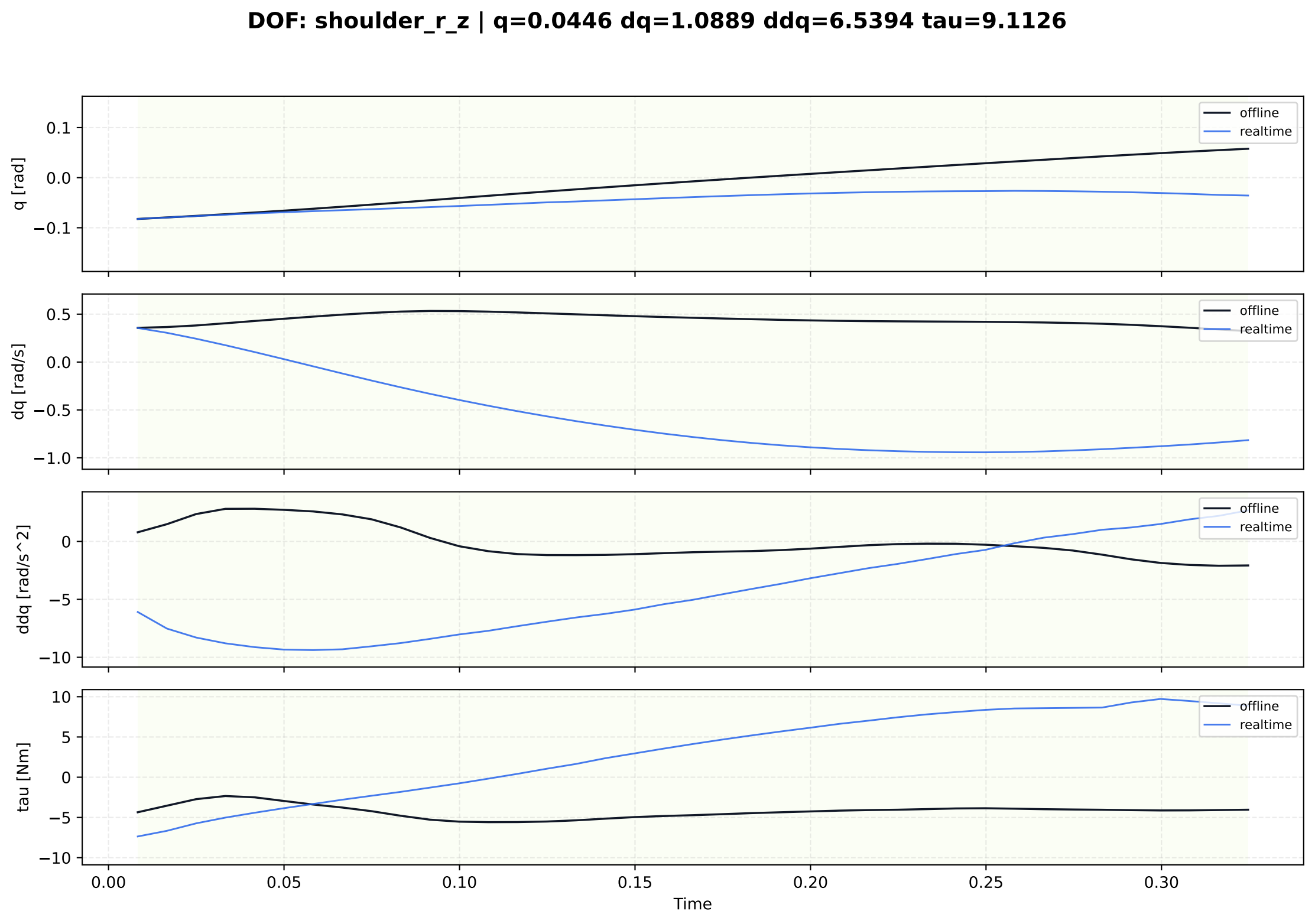

DOF: scapula_abduction_l | q=0.0114 dq=0.1167 ddq=1.9361 tau=4.2585
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

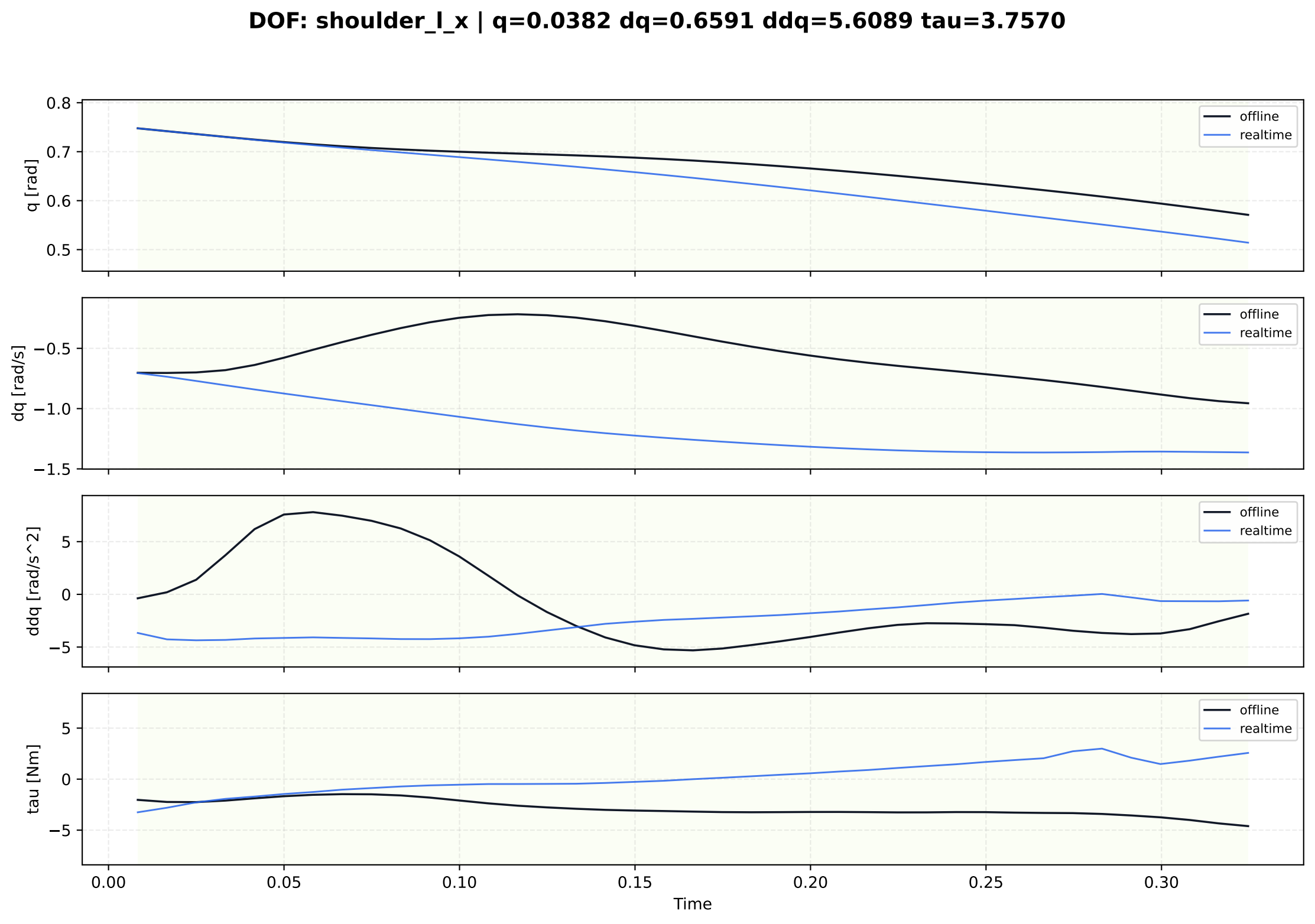

DOF: scapula_elevation_l | q=0.0148 dq=0.7774 ddq=4.3408 tau=5.4403
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

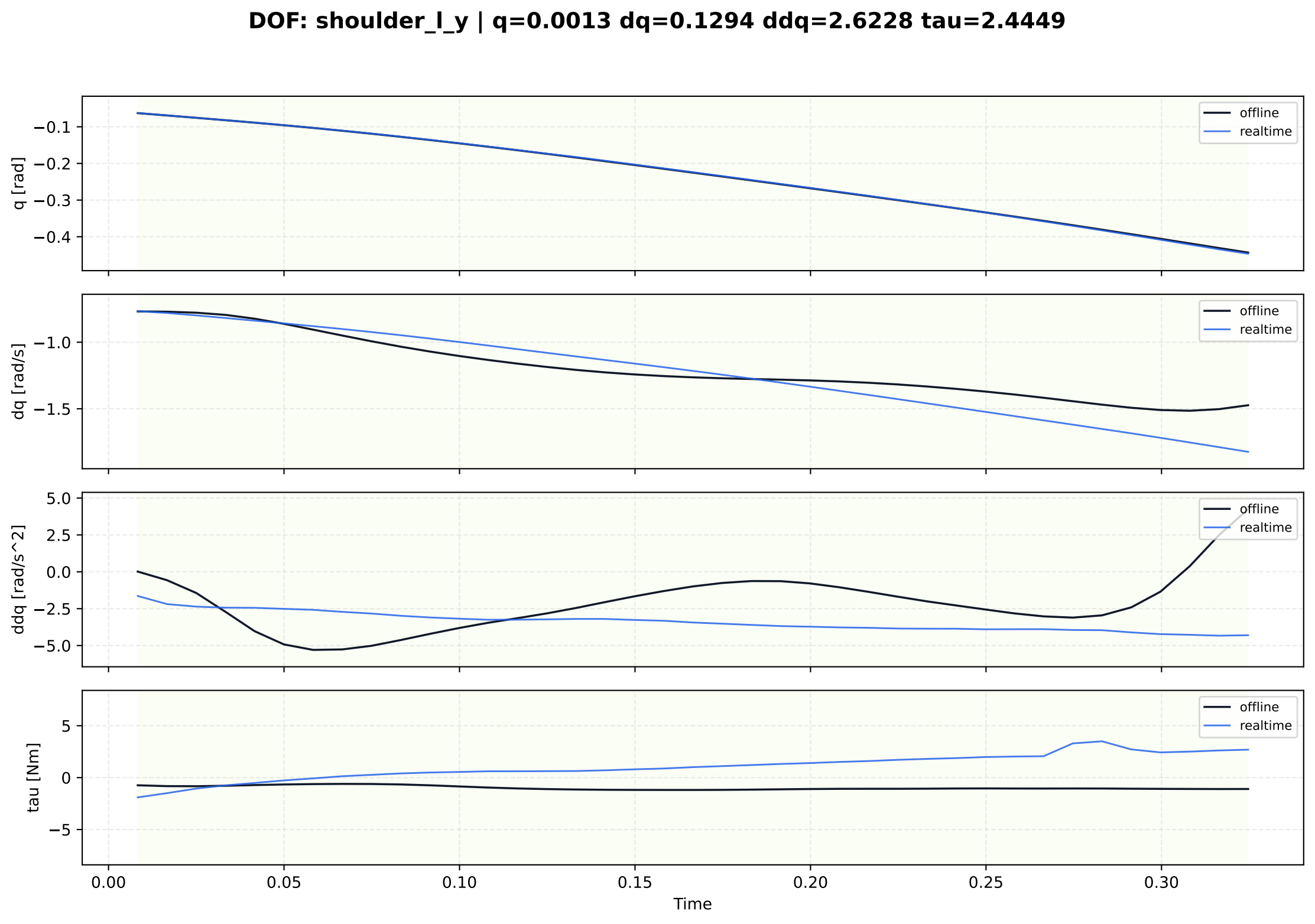

DOF: scapula_upward_rot_l | q=0.0399 dq=0.6604 ddq=4.3980 tau=4.9558
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

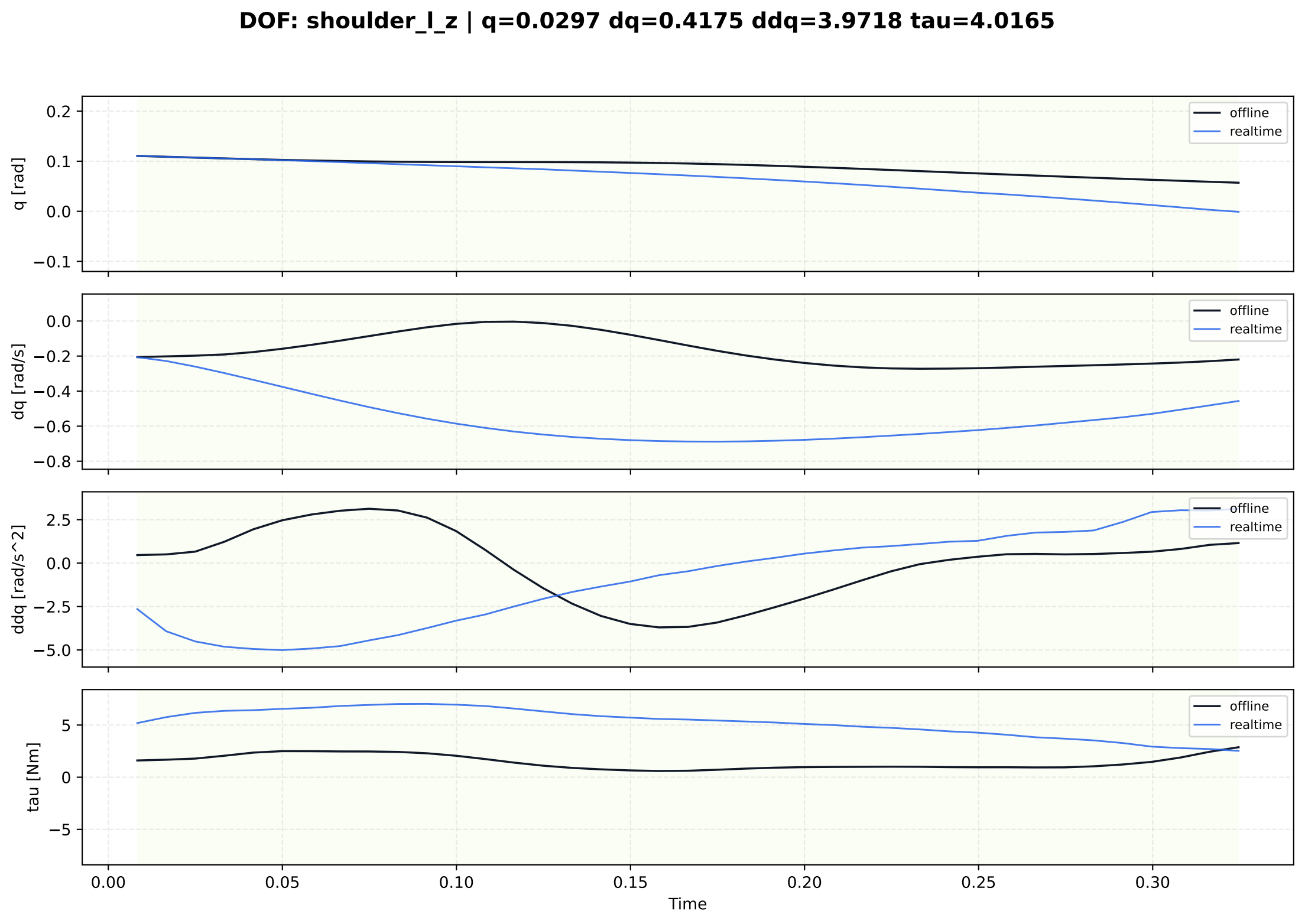

DOF: shoulder_r_x | q=0.0511 dq=0.7725 ddq=7.5071 tau=6.6900
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

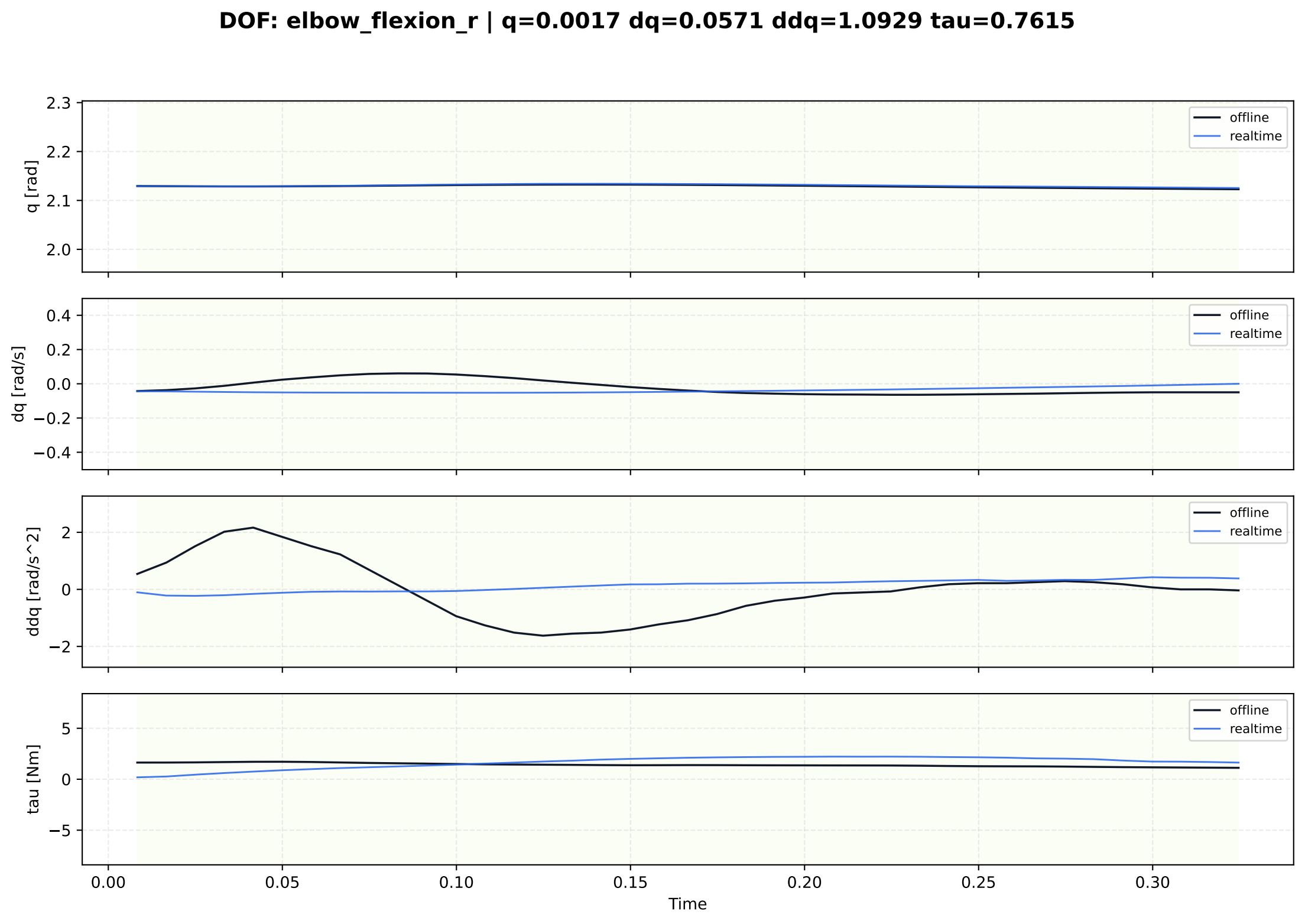

DOF: shoulder_r_y | q=0.0087 dq=0.1422 ddq=1.4265 tau=3.2040
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
Time
offline
realtime

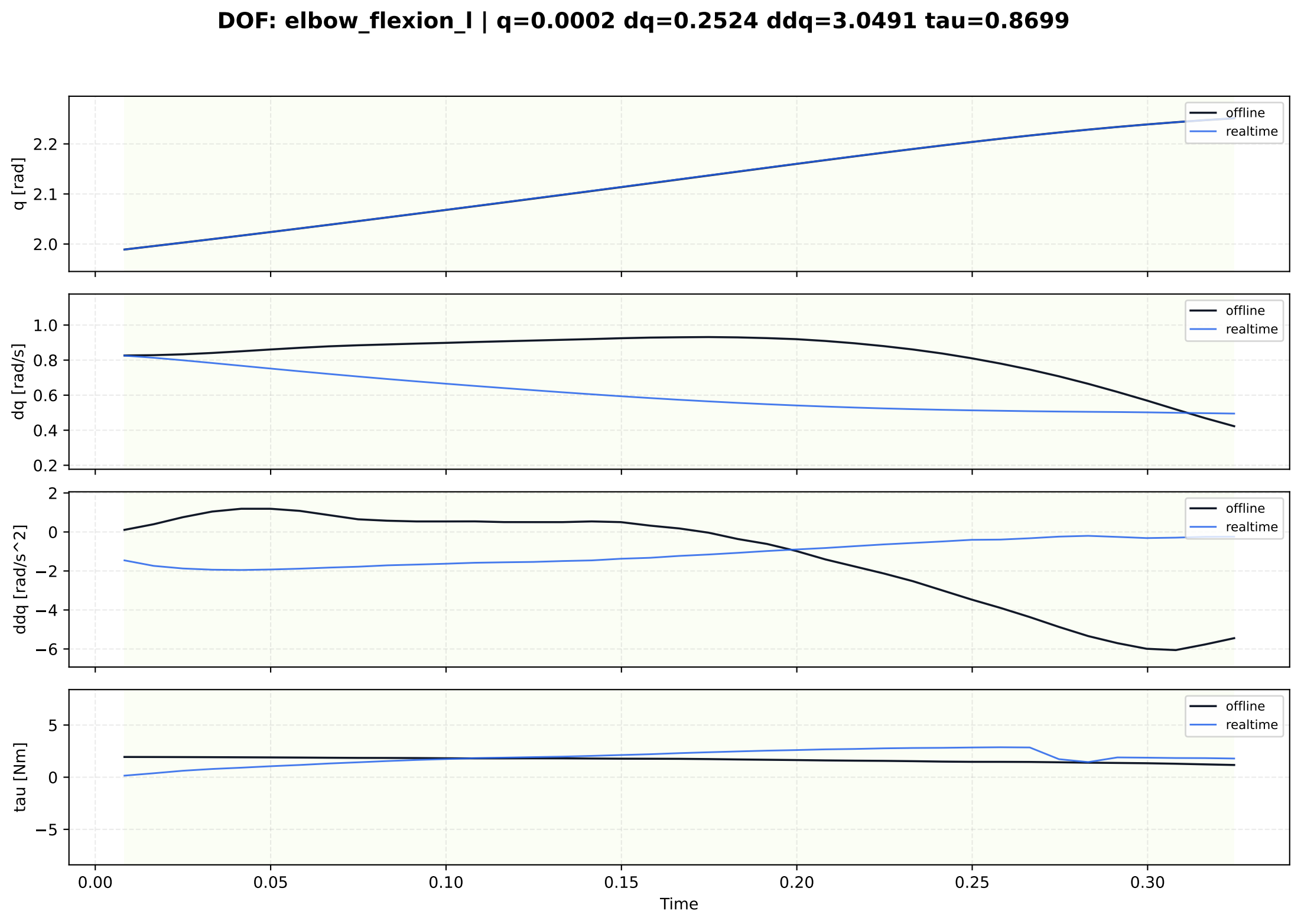

DOF: shoulder_r_z | q=0.0446 dq=1.0889 ddq=6.5394 tau=9.1126
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

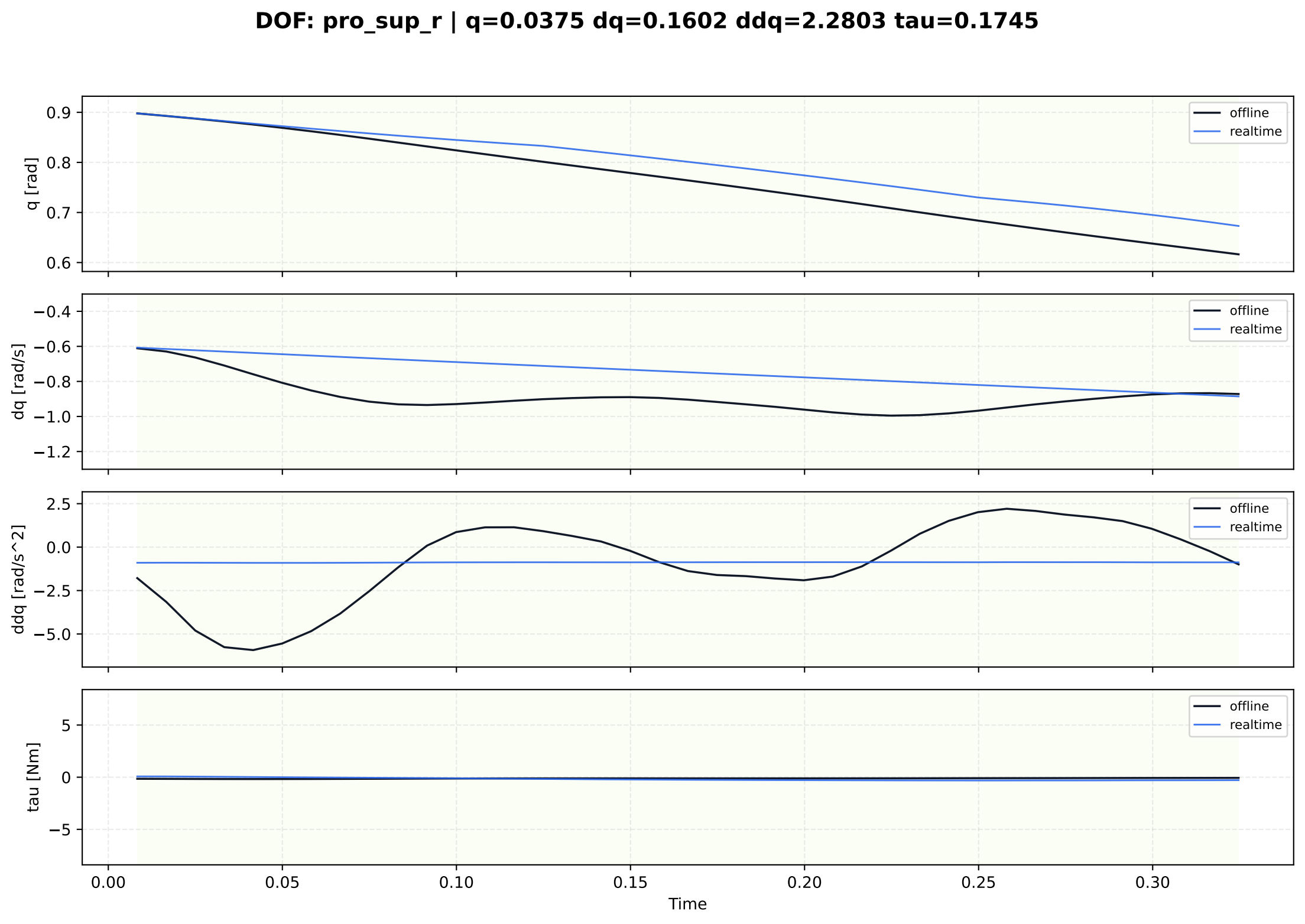

DOF: shoulder_l_x | q=0.0382 dq=0.6591 ddq=5.6089 tau=3.7570
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

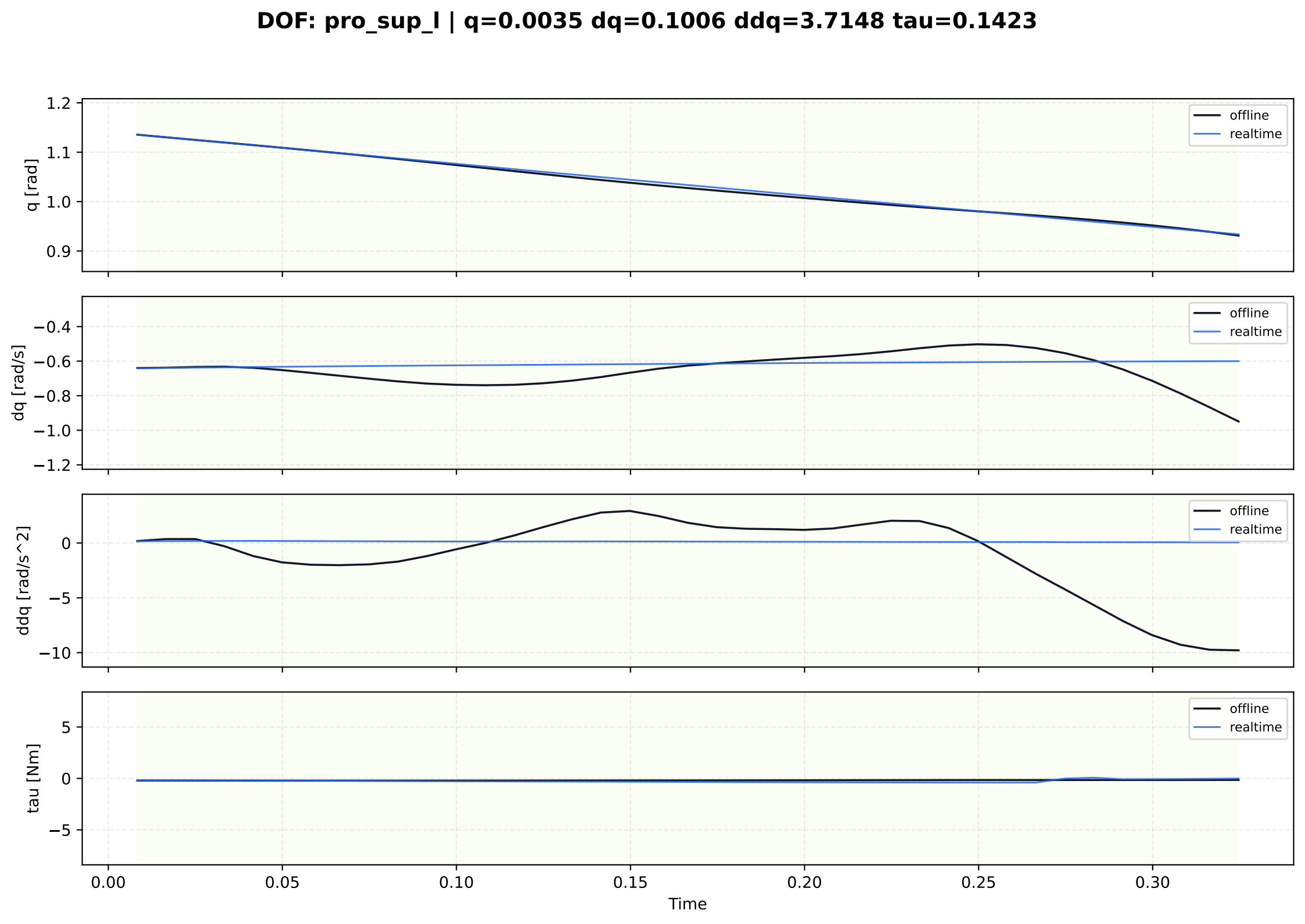

DOF: shoulder_l_y | q=0.0013 dq=0.1294 ddq=2.6228 tau=2.4449
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

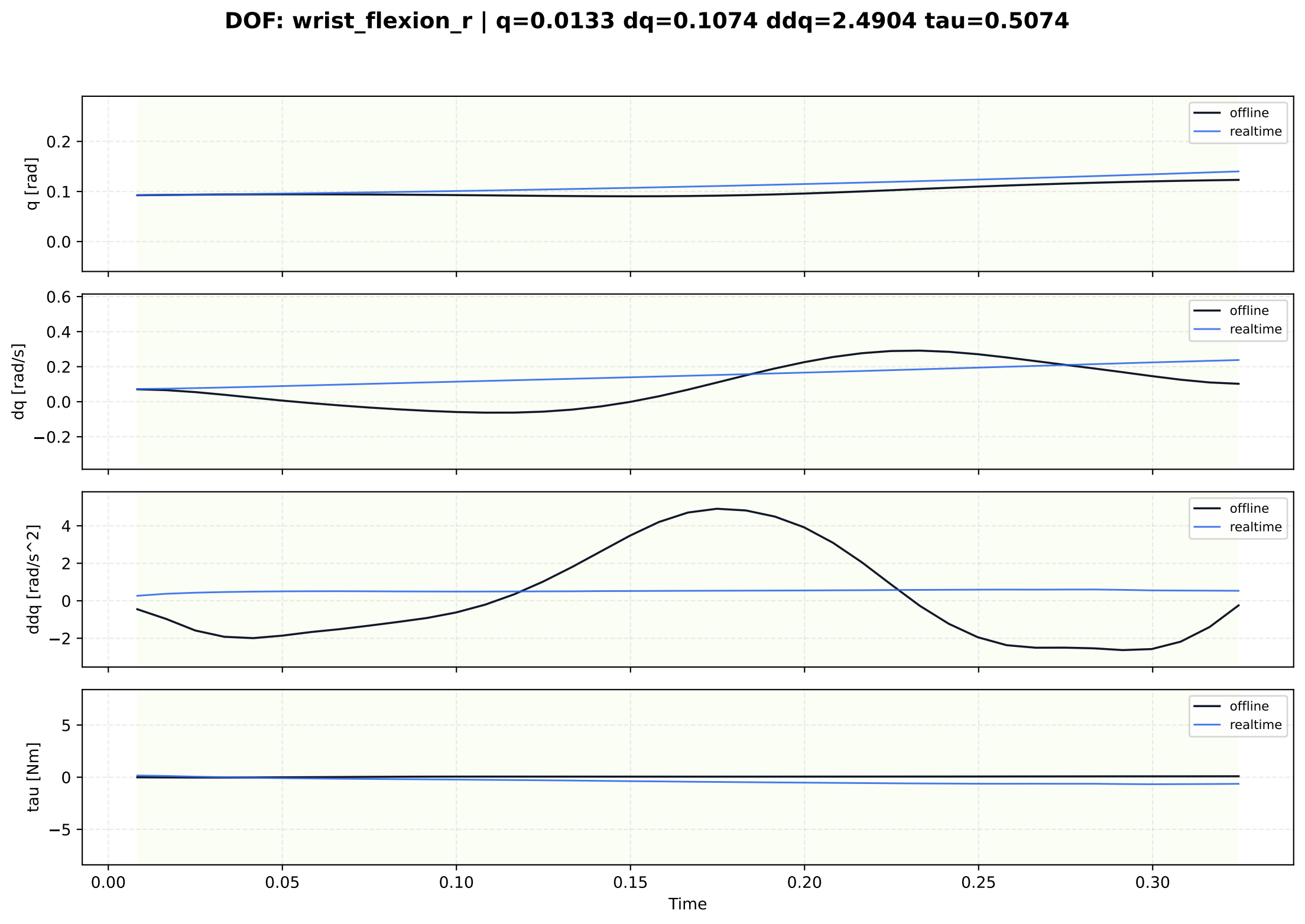

DOF: shoulder_l_z | q=0.0297 dq=0.4175 ddq=3.9718 tau=4.0165
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

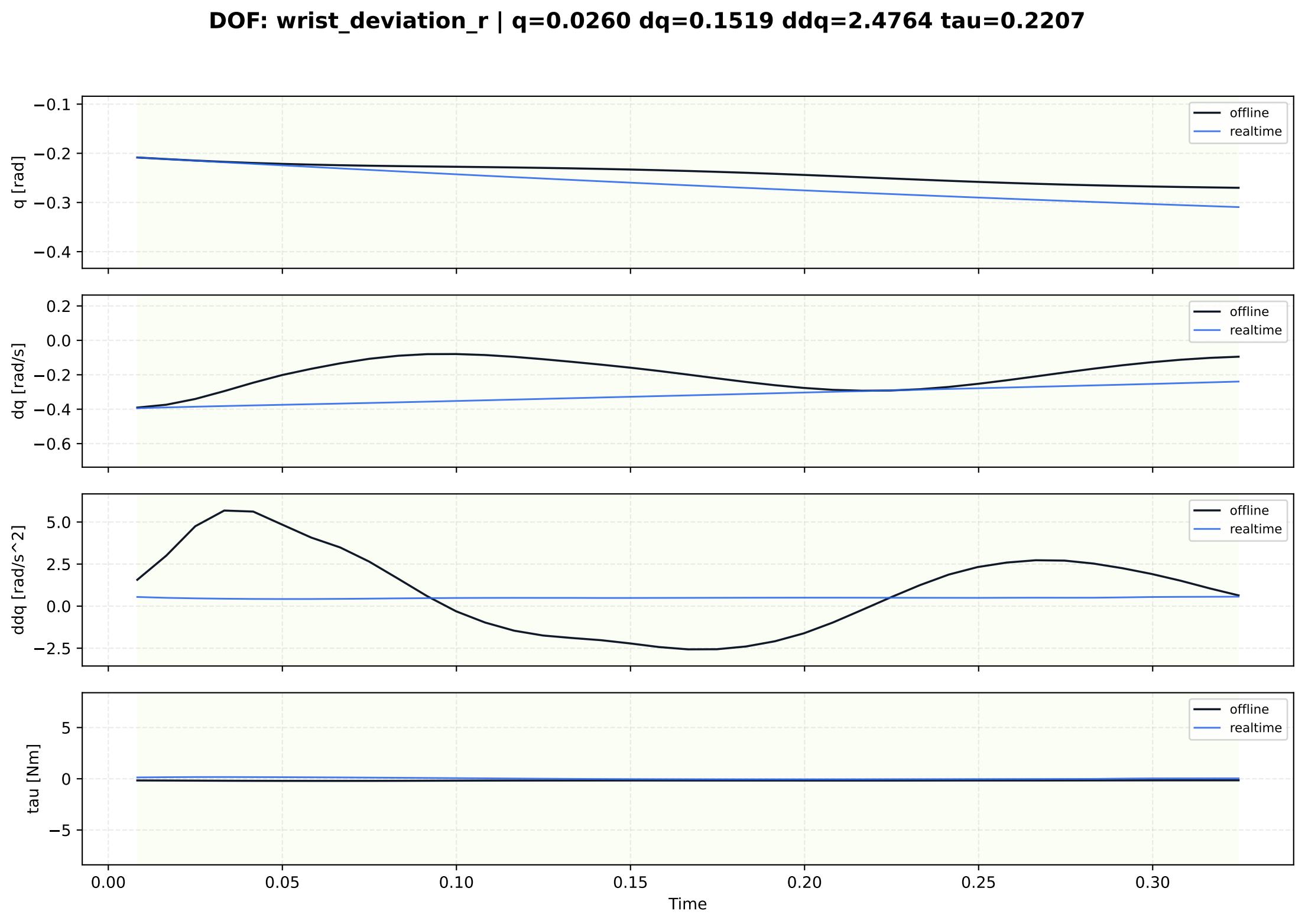

DOF: elbow_flexion_r | q=0.0017 dq=0.0571 ddq=1.0929 tau=0.7615
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

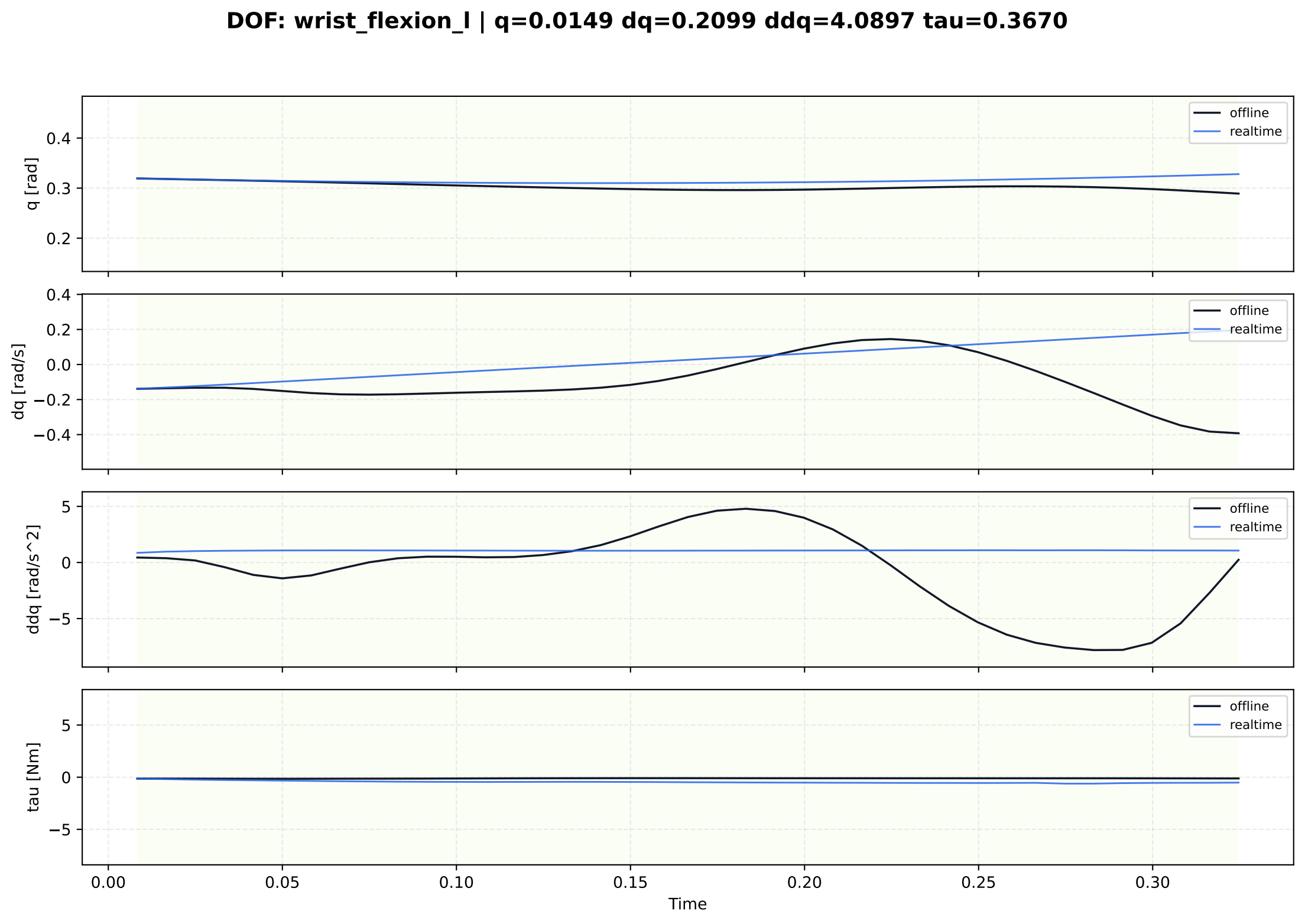

DOF: elbow_flexion_l | q=0.0002 dq=0.2524 ddq=3.0491 tau=0.8699
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

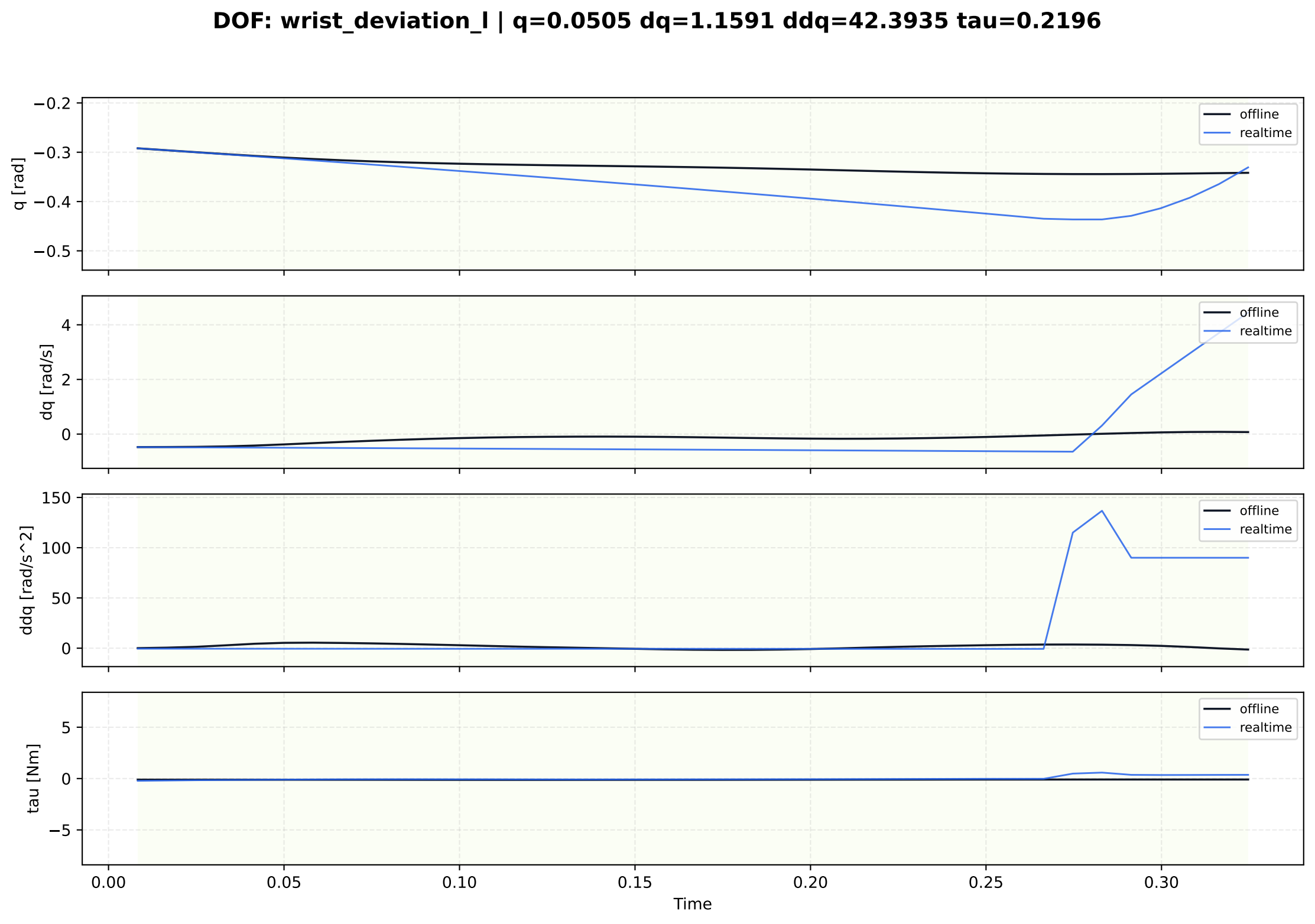

DOF: pro_sup_r | q=0.0375 dq=0.1602 ddq=2.2803 tau=0.1745
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
Time
offline
realtime

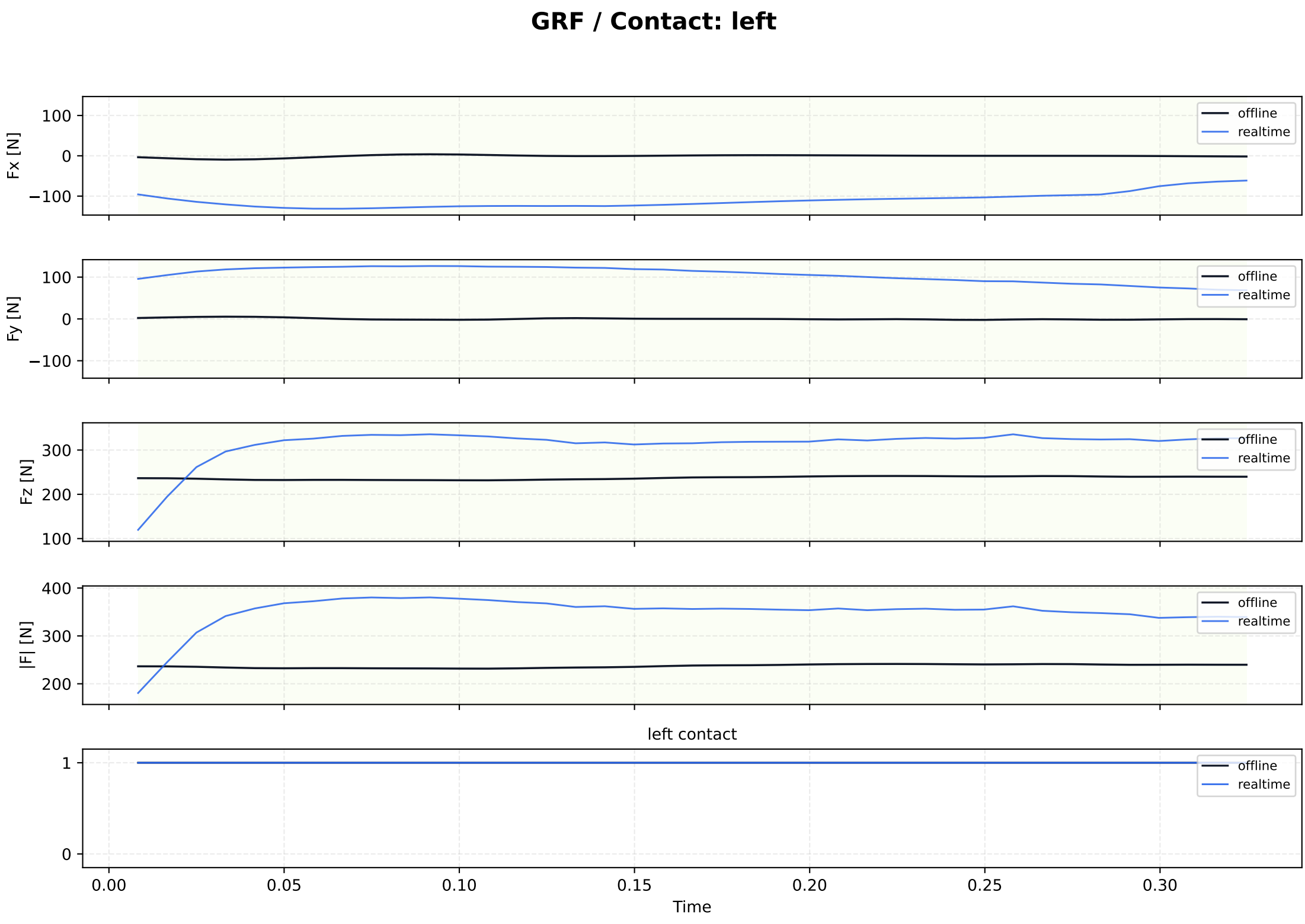

DOF: pro_sup_l | q=0.0035 dq=0.1006 ddq=3.7148 tau=0.1423
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time

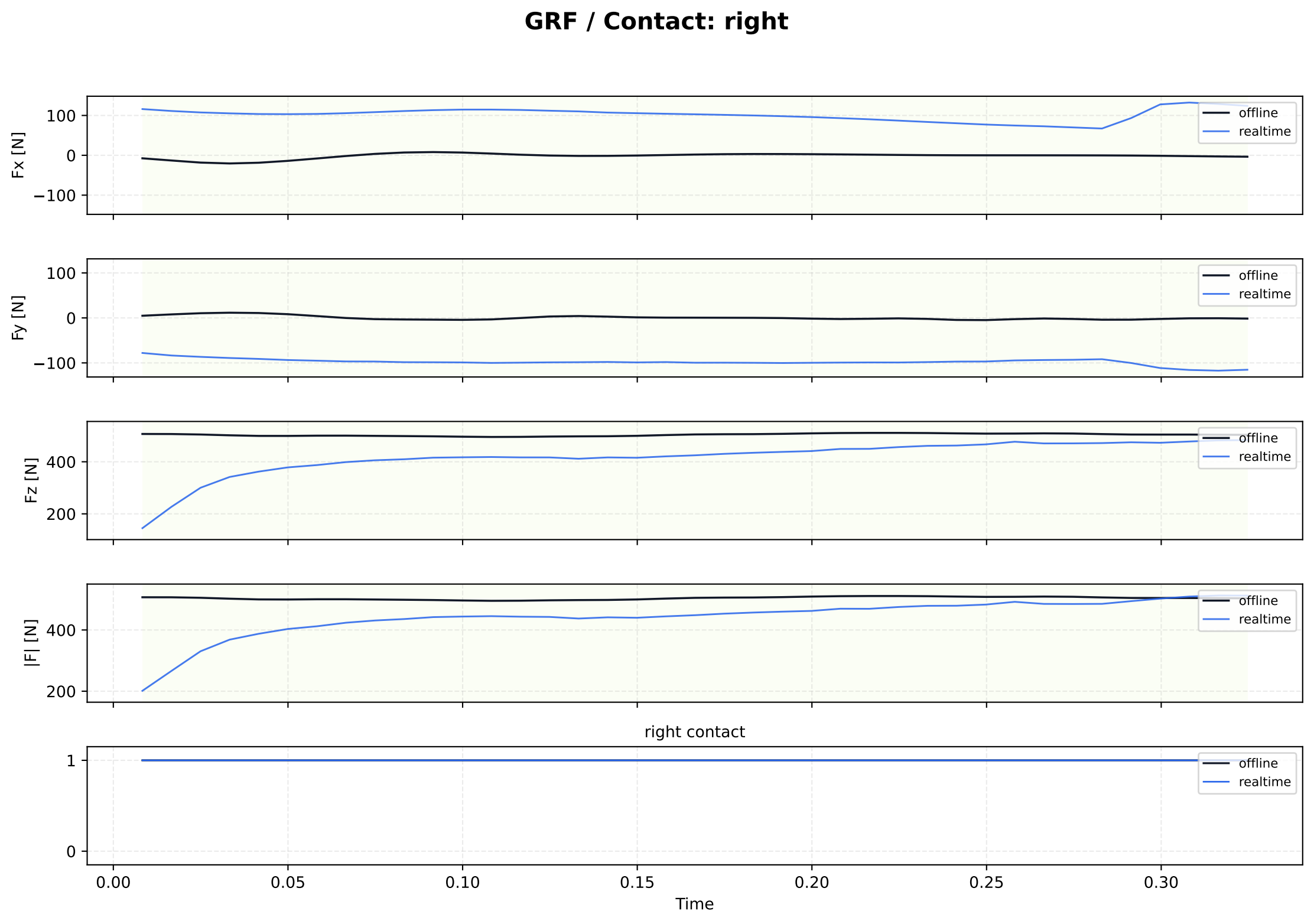

DOF: wrist_flexion_r | q=0.0133 dq=0.1074 ddq=2.4904 tau=0.5074
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time


DOF: wrist_deviation_r | q=0.0260 dq=0.1519 ddq=2.4764 tau=0.2207
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time


DOF: wrist_flexion_l | q=0.0149 dq=0.2099 ddq=4.0897 tau=0.3670
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time


DOF: wrist_deviation_l | q=0.0505 dq=1.1591 ddq=42.3935 tau=0.2196
q [rad]
dq [rad/s]
ddq [rad/s^2]
tau [Nm]
offline
realtime
Time


GRF / Contact: left
Fx [N]
Fy [N]
Fz [N]
|F| [N]
offline
realtime
left contact
Time


GRF / Contact: right
Fx [N]
Fy [N]
Fz [N]
|F| [N]
offline
realtime
right contact
Time
0.00
0.05
0.10
0.15
0.20
0.25
0.30